

TECHNIQUES IN EXOTICA

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ABSTRACT. IAS/PCMI lecture notes.

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¹Based on lectures by Tye Lidman and Lisa Piccirillo [21].

Part 1. Exotic Manifolds²

1. EXOTICA FROM SEIBERG–WITTEN THEORY

All manifolds are connected, oriented, smooth, and closed unless a boundary is displayed. Homology has $\mathbb{Z}$ -coefficients, and

$$Q_X: H_2(X) \times H_2(X) \rightarrow \mathbb{Z}, \quad \sigma(X) = b^+(X) - b^-(X), \quad H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$\overline{\mathbb{CP}}^2 = -\mathbb{CP}^2, \quad \#_k M = \#_{i=1}^k M.$$

1.1. Formal Structure and Seiberg–Witten Obstructions. For a cobordism $W: Y_0 \rightarrow Y_1$, write $Y_0 \simeq_h Y_1$ when both inclusions $Y_i \hookrightarrow W$ are homotopy equivalences. A product h -cobordism identifies its ends, while diffeomorphic ends need not make a prescribed h -cobordism a product.

Definition 1.1.1 (Exotic manifolds and surfaces, [21]). A four-manifold X and a surface $S \subseteq X$ are *exotic* when

$$\begin{aligned} X &\cong_{\text{homeo}} X', & X &\not\cong_{\text{sm}} X', \\ (X, S) &\cong_{\text{homeo}} (X, S'), & (X, S) &\not\cong_{\text{sm}} (X, S'). \end{aligned}$$

The operation–auxiliary–obstruction scheme is

$$\begin{array}{ccc} (X, S) & \xrightarrow{\text{operation}} & (X, S') \\ \text{auxiliary} \downarrow & & \downarrow \text{auxiliary} \\ Z_{X,S} & \xrightarrow{\text{SW, HF}^-} & Z_{X,S'} \end{array} \quad Z_{X,S} \in \{X \setminus \text{int}(\nu S), \Sigma_2(X, S)\}.$$

Theorem 1.1.2 (Classification, cobordism, and stability, [32, 36, 10, 11, 18]). *For simply connected closed oriented four-manifolds,*

$$Q_M \cong Q_{M'} \iff M \cong_{\text{homeo}} M', \quad M \simeq M' \implies M \simeq_h M'.$$

Every form isometry is realized, and its realization is unique up to topological isotopy:

$$f_* = g_*: H_2(M) \longrightarrow H_2(M') \implies f \simeq_{\text{isot.top}} g.$$

For a smooth h -cobordism with simply connected ends of dimension at least five,

$$W: Y_0 \rightarrow Y_1 \implies W \cong_{\text{sm}} Y_0 \times I \text{ rel } Y_0.$$

If X, X' are also indefinite and $\phi: Q_X \rightarrow Q_{X'}$ is an isometry, then for $k \gg 0$,

$$\Phi: X \#_k (S^2 \times S^2) \xrightarrow{\cong_{\text{sm}}} X' \#_k (S^2 \times S^2), \quad \Phi_* = \phi \oplus \text{id}_{kH}.$$

Fact 1.1.3 (Dimensional contrast, [26, 36]).

$$\begin{aligned} \Sigma^7 &\cong_{\text{homeo}} S^7, & \Sigma^7 &\not\cong_{\text{sm}} S^7, \\ X^4 &\cong_{\text{homeo}} X'^4, & X^4 &\not\cong_{\text{sm}} X'^4 \xrightarrow{1.1.2} X \simeq_h X' \quad \text{through no product } h\text{-cobordism.} \end{aligned}$$

Proposition 1.1.4 (Stable h -cobordism, [36, 21]). *Homeomorphic simply connected, closed, oriented, indefinite four-manifolds are smoothly h -cobordant.*

²Based on lectures by Tye Lidman and Lisa Piccirillo [21].

Proof. Attach and cancel the hyperbolic stabilizations supplied by Wall realization.

$$\begin{array}{c}
 X \cong_{\text{homeo}} X' \xrightarrow{1.1.2} Q_X \cong Q_{X'} \xrightarrow{1.1.2} \Phi: X\#_k(S^2 \times S^2) \cong_{\text{sm}} X'\#_k(S^2 \times S^2). \\
 \begin{array}{ccccc}
 X & \xleftarrow{W_X} & X\#_k(S^2 \times S^2) & \xrightarrow{\Phi} & X'\#_k(S^2 \times S^2) & \xrightarrow{\quad} & X' \\
 & & \searrow & & \downarrow & & \swarrow \\
 & & & & W_X \cup_{\Phi} \overline{W_{X'}} & &
 \end{array}
 \end{array}$$

$$W := W_X \cup_{\Phi} \overline{W_{X'}}, \quad \pi_1(W) = 1, \quad H_*(X) \xrightarrow{\cong} H_*(W) \xleftarrow{\cong} H_*(X').$$

Van Kampen, Mayer–Vietoris, and Whitehead make both end inclusions homotopy equivalences, so W is the required smooth h -cobordism. $\square$

Lemma 1.1.5 (Surgery on a product sphere, [21]). *Surgery on $S^2 \times \{\text{pt}\} \subseteq X\#(S^2 \times S^2)$ returns X .*

Proof. The surgery replaces one standard piece and converts the stabilizing summand into S^4 .

$$\begin{aligned}
 S^2 \times S^2 &= (S^2 \times D^2) \cup (S^2 \times D^2) \xrightarrow{S^2 \times D^2 \rightsquigarrow D^3 \times S^1} S^4 = (S^2 \times D^2) \cup (D^3 \times S^1) \\
 X\#(S^2 \times S^2) &\xrightarrow{\text{surgery}} X\#S^4 \xrightarrow{\cong} X.
 \end{aligned}$$

Thus product-sphere surgery cancels one $S^2 \times S^2$ stabilization. $\square$

Theorem 1.1.6 (Stable exotic sphere links, [21]). *If X is simply connected, closed, and exotic, then for $k \gg 0$, $X\#_k(S^2 \times S^2)$ contains an exotic link*

$$S = S_1 \sqcup \cdots \sqcup S_k, \quad S_i \cong S^2.$$

Proof. Transport the standard product-sphere link across a stable diffeomorphism to an exotic copy X' of X .

$$\begin{aligned}
 L_X &:= \bigsqcup_{i=1}^k (S^2 \times \{\text{pt}\})_i, & M &:= X'\#_k(S^2 \times S^2), & \Phi: X\#_k(S^2 \times S^2) &\cong_{\text{sm}} M. \\
 S &:= \Phi(L_X), & L_{X'} &:= \bigsqcup_{i=1}^k (S^2 \times \{\text{pt}\})_i, & (M, S) &\cong_{\text{homeo}} (M, L_{X'}). \\
 \begin{array}{ccc}
 (M, S) & \xrightarrow{\cong_{\text{homeo}}} & (M, L_{X'}) \\
 \text{surgery} \downarrow & & \downarrow \text{surgery} \\
 X & \xrightarrow{\cong_{\text{sm}}} & X'
 \end{array}
 \end{aligned}$$

By Lemma 1.1.5, a diffeomorphism of the upper pairs would descend to $X \cong_{\text{sm}} X'$, so the sphere link is exotic. $\square$

Together with Theorem 1.1.13, Theorem 1.1.6 produces exotic surfaces [6, 21].

Proposition 1.1.7 (Exceptional-sphere transfer, [21]). *Suppose X, X' are closed, simply connected, and exotic, and*

$$f: X \xrightarrow{\cong_{\text{homeo}}} X', \quad g: X\#\overline{\mathbb{CP}}^2 \xrightarrow{\cong_{\text{sm}}} X'\#\overline{\mathbb{CP}}^2, \quad g_* = f_* \oplus \text{id}.$$

Then $X\#\overline{\mathbb{CP}}^2$ contains an exotic sphere.

Proof. Compare the standard exceptional sphere with the pullback of the exceptional sphere in the exotic blow-up.

$$\begin{array}{ccc}
 M := X \# \overline{\mathbb{CP}}^2, & E \subset M, & E' \subset X' \# \overline{\mathbb{CP}}^2, & S := g^{-1}(E'). \\
 (M, E) \cong_{\text{homeo}} (M, S) & [11]. & & \\
 (M, E) \xrightarrow{\cong_{\text{homeo}}} (M, S) & & & \\
 \downarrow \text{blow down} & & \downarrow \text{blow down} & \\
 X \xrightarrow{\not\cong_{\text{sm}}} X' & & &
 \end{array}$$

A diffeomorphism of the upper pairs would descend to a diffeomorphism of the lower manifolds, so E and S are exotic. $\square$

Witten's invariant and its basic set are [37, 28, 30]

$$\text{SW}_X: \text{Spin}^c(X) \rightarrow \mathbb{Z}, \quad \mathcal{B}(X) := \{\mathfrak{s} \in \text{Spin}^c(X) \mid \text{SW}_X(\mathfrak{s}) \neq 0\}.$$

The affine action, characteristic classes, and sign assumptions are

$$\mathfrak{s} + a \in \text{Spin}^c(X), \quad c_1(\mathfrak{s} + a) = c_1(\mathfrak{s}) + 2a, \quad c_1(\mathfrak{s}) \equiv w_2(X) \pmod{2}, \quad b^+(X) \geq 2.$$

The remaining conventions are

$$\pi_1(X) = 1 \implies c_1: \text{Spin}^c(X) \xrightarrow{\cong} \{c \in H^2(X; \mathbb{Z}) \mid c \equiv w_2(X) \pmod{2}\}.$$

$$\pi_1(X) \neq 1 \implies c_1 \text{ need not be injective or surjective.}$$

$$0 \in \text{Spin}^c(X) \text{ only for spin } X, \quad \text{a homology orientation fixes the sign of } \text{SW}_X.$$

Theorem 1.1.8 (Seiberg–Witten package, [37, 19, 28, 30, 35]). *Let $b^+(X) \geq 2$. Naturality and finiteness give*

$$f: X \xrightarrow{\cong_{\text{sm}}} X' \implies \text{SW}_X(\mathfrak{s}) = \text{SW}_{X'}(f_*\mathfrak{s}), \quad \#\mathcal{B}(X) < \infty.$$

For $\mathfrak{s} \in \mathcal{B}(X)$ and a connected, oriented, non-torsion surface $F \subseteq X$ with $F^2 \geq 0$,

$$(\text{SW-A}) \quad |\langle c_1(\mathfrak{s}), [F] \rangle| + F^2 \leq 2g(F) - 2.$$

Vanishing and symplectic nonvanishing are

$$X = X_1 \# X_2, \quad b^+(X_1)b^+(X_2) > 0 \implies \text{SW}_X \equiv 0,$$

$$X \text{ admits a metric with positive scalar curvature} \implies \text{SW}_X \equiv 0.$$

$$(X, \omega) \text{ symplectic} \implies \text{SW}_X(\mathfrak{s}_\omega) = \pm 1.$$

For the exceptional class E ,

$$\langle c_1(\mathfrak{s}_\pm), E \rangle = \pm 1, \quad \text{SW}_{X \# \overline{\mathbb{CP}}^2}(\mathfrak{s} \# \mathfrak{s}_\pm) = \text{SW}_X(\mathfrak{s}),$$

$$\mathfrak{s} \in \mathcal{B}(X) \iff \mathfrak{s} \# \mathfrak{s}_\pm \in \mathcal{B}(X \# \overline{\mathbb{CP}}^2).$$

For simple-type X , these are all basic structures on the blow-up.

Fact 1.1.9 (Moduli dimension, simple type, and wall crossing, [37, 28, 30, 15]). *Let $\mathcal{M}(X, \mathfrak{s}, g, \eta)$ be the perturbed monopole moduli space. For $b_1(X) = 0$,*

$$d(\mathfrak{s}) = \frac{c_1(\mathfrak{s})^2 - 3\sigma(X) - 2\chi(X)}{4}, \quad \text{SW}_X(\mathfrak{s}) \neq 0 \implies d(\mathfrak{s}) \in 2\mathbb{Z}_{\geq 0}.$$

If $d(\mathfrak{s}) = 0$, then

$$c_1(\mathfrak{s})^2 = 3\sigma(X) + 2\chi(X), \quad \text{SW}_X(\mathfrak{s}) = \#\mathcal{M}(X, \mathfrak{s}, g, \eta).$$

The simple-type conjecture and the $b^+ = 1$ chamber formula are

$$\begin{aligned} b^+(X) &\geq 2, & d(\mathfrak{s}) > 0 &\xrightarrow{\text{conj.}} \text{SW}_X(\mathfrak{s}) = 0, \\ b_1(X) &= 0, & b^+(X) &= 1, & d(\mathfrak{s}) = 0 &\implies \text{SW}_X^+(\mathfrak{s}) - \text{SW}_X^-(\mathfrak{s}) = \pm 1. \\ X &= \mathbb{CP}^2, & c_1(\mathfrak{s}) &= 3H &\implies \{\text{SW}_{\mathbb{CP}^2}^+(\mathfrak{s}), \text{SW}_{\mathbb{CP}^2}^-(\mathfrak{s})\} &= \{0, \pm 1\}. \end{aligned}$$

Example 1.1.10 (The $K3$ surface, [28]).

$$Q_{K3} \cong 2(-E_8) \oplus 3H, \quad c_1(\mathfrak{s}_0) = 0, \quad \text{SW}_{K3}(\mathfrak{s}_0) = \pm 1, \quad \mathcal{B}(K3) = \{\mathfrak{s}_0\}.$$

For $k \geq 1$,

$$\begin{aligned} c_1\left(\mathcal{B}(K3 \#_k \overline{\mathbb{CP}}^2)\right) &= \{\varepsilon_1 E_1 + \cdots + \varepsilon_k E_k \mid \varepsilon_i \in \{\pm 1\}\}, \\ \#\mathcal{B}(K3 \#_k \overline{\mathbb{CP}}^2) &= 2^k, \quad \text{SW}_{K3 \# \overline{\mathbb{CP}}^2} \not\equiv 0, \quad \text{SW}_{\overline{K3} \# \overline{\mathbb{CP}}^2} \equiv 0. \end{aligned}$$

Theorem 1.1.11 (Donaldson diagonalization, [5]). *If X is closed, smooth, and simply connected and Q_X is definite, then*

$$Q_X \cong b_2(X)\langle 1 \rangle \quad \text{or} \quad Q_X \cong b_2(X)\langle -1 \rangle.$$

Proposition 1.1.12 (Adjunction consequences, [5, 37, 28, 35]). *The $K3$ surface contains no square-2 torus, and*

$$K3 \cong_{\text{sm}} X_1 \# X_2 \implies X_1 \cong_{\text{homeo}} S^4 \quad \text{or} \quad X_2 \cong_{\text{homeo}} S^4.$$

The coordinate tori span $H_2(T^4)$, and

$$\mathcal{B}(T^4) = \{\mathfrak{s}_0\}, \quad c_1(\mathfrak{s}_0) = 0, \quad \text{SW}_{T^4}(\mathfrak{s}_0) = \pm 1.$$

If X, X' are closed symplectic and $\pi_1(Y) \rightarrow G \neq 1$ for finite G , then $X \# X'$ and $X \# Y$ are not symplectic.

Proof. Apply adjunction and connected-sum vanishing to the unique basic structure of $K3$.

$$F \cong T^2, \quad F^2 = 2, \quad c_1(\mathfrak{s}_0) = 0, \quad \text{SW}_{K3}(\mathfrak{s}_0) \neq 0 \xrightarrow{\text{SW-A}} 2 \leq 0.$$

$$K3 = X_1 \# X_2, \quad b^+(X_1)b^+(X_2) > 0 \xrightarrow{1.1.8} \text{SW}_{K3} = 0.$$

$$b^+(X_2) = 0, \quad Q_{X_2} \text{ even definite} \xrightarrow{1.1.11} Q_{X_2} = 0 \xrightarrow{1.1.2} X_2 \cong_{\text{homeo}} S^4.$$

The coordinate tori annihilate every basic first Chern class.

$$\text{SW}_{T^4}(\mathfrak{s}_0) = \pm 1, \quad \mathfrak{s} \in \mathcal{B}(T^4) \xrightarrow{\text{SW-A}} \langle c_1(\mathfrak{s}), [T_{ij}] \rangle = 0 \quad (1 \leq i < j \leq 4),$$

$$\langle [T_{ij}] \rangle = H_2(T^4), \quad H^2(T^4) \text{ torsion-free} \implies c_1(\mathfrak{s}) = 0 \implies \mathfrak{s} = \mathfrak{s}_0.$$

Finally, Taubes contradicts connected-sum vanishing, directly and on the finite cover induced by G .

$$[\omega]^2 > 0 \implies b^+(X), b^+(X') \geq 1 \xrightarrow{1.1.8} \text{SW}_{X \# X'} = 0 \neq \text{SW}_{(X \# X', \omega)}(\mathfrak{s}_\omega),$$

$$\widetilde{X \# Y} \cong \widetilde{Y} \#_{|G|} X, \quad |G| > 1 \xrightarrow{1.1.8} \text{SW}_{\widetilde{X \# Y}} = 0 \neq \text{SW}_{(\widetilde{X \# Y}, \widetilde{\omega})}(\mathfrak{s}_{\widetilde{\omega}}).$$

Thus the three assertions follow. $\square$

Theorem 1.1.13 (Donaldson's first exotica, [6]). *There exist simply connected, closed, exotic four-manifolds.*

Proof. Compare the blown-up $K3$ with the standard manifold having the same odd form.

$$X := K3 \# \overline{\mathbb{CP}}^2, \quad X' := \#_3 \mathbb{CP}^2 \#_{20} \overline{\mathbb{CP}}^2.$$

$$Q_X = 2(-E_8) \oplus 3H \oplus \langle -1 \rangle \cong 3\langle 1 \rangle \oplus 20\langle -1 \rangle \cong Q_{X'} \xrightarrow{1.1.2} X \cong_{\text{homeo}} X'.$$

$$\text{SW}_{K3}(\mathfrak{s}_0) = \pm 1 \xrightarrow{1.1.8} \text{SW}_X \neq 0, \quad X' = \mathbb{CP}^2 \# (\#_2 \mathbb{CP}^2 \#_{20} \overline{\mathbb{CP}}^2) \xrightarrow{1.1.8} \text{SW}_{X'} \equiv 0.$$

Naturality gives $X \not\cong_{\text{sm}} X'$, so (X, X') is exotic. $\square$

Theorem 1.1.14 (Negative torus-knot trace obstruction, [21]). *Let $X_n(K)$ be B^4 with an n -framed 2-handle attached along $K \subseteq S^3$. If K is a nontrivial negative torus knot and $\text{SW}_M \neq 0$, then*

$$n \geq 0 \implies X_n(K) \not\hookrightarrow_{\text{sm}} M.$$

Proof. By Figure 1, blow up a coherent negative-crossing unknotting sequence and apply adjunction to the capped core.

$$X_n(K) \hookrightarrow M \implies X_{n+4u}(U) \hookrightarrow M \#_u \overline{\mathbb{CP}}^2, \quad u > 0, \quad n + 4u > 0.$$

$$S := \text{capped handle core}, \quad S \cong S^2, \quad S^2 = n + 4u.$$

$$\text{SW}_M \neq 0 \xrightarrow{1.1.8} \text{SW}_{M \#_u \overline{\mathbb{CP}}^2} \neq 0 \xrightarrow{\text{SW-A}} |\langle c_1(\mathfrak{s}), [S] \rangle| + n + 4u \leq -2.$$

This contradicts $n + 4u > 0$, so the trace cannot embed. $\square$

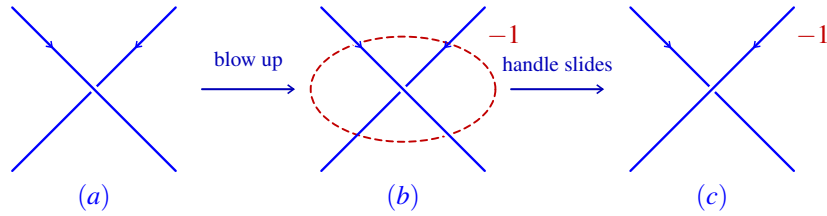


FIGURE 1. A blow-up and handle slides switch a negative crossing.

Theorem 1.1.15 ($-E_8$ -replacement obstruction, [21]). *Let P_{-E_8} be the simply connected negative E_8 -plumbing with*

$$Y = \partial P_{-E_8} = S_{-1}^3(T_{-2,3}),$$

and let Z complement two disjoint copies of P_{-E_8} in $K3$. Then

$$\mathcal{Q}_Z \cong \bigoplus_{i=1}^3 \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix}.$$

Choose sphere–torus bases (S_i, T_i) for these summands, so

$$S_i^2 = -2, \quad T_i^2 = 0, \quad S_i \cdot T_i = 1.$$

The replacement

$$K3 = Z \cup_Y P_{-E_8} \cup_Y P_{-E_8} \xrightarrow{P_{-E_8} \rightsquigarrow X_{-1}(T_{-2,3})} X^* := Z \cup_Y P_{-E_8} \cup_Y X_{-1}(T_{-2,3})$$

satisfies $\mathcal{B}(X^) = \emptyset$.*

Proof. The dimension formula and adjunction impose incompatible squares on a basic class.

$$\sigma(X^*) = -9, \quad \chi(X^*) = 17, \quad d(\mathfrak{s}) = \frac{c_1(\mathfrak{s})^2 - 7}{4}, \quad \mathfrak{s} \in \mathcal{B}(X^*) \implies c_1(\mathfrak{s})^2 \geq 7.$$

$$T_i^2 = (T_i + S_i)^2 = 0, \quad g(T_i) = g(T_i + S_i) = 1 \xrightarrow{\text{SW-A}} \langle c_1(\mathfrak{s}), T_i \rangle = \langle c_1(\mathfrak{s}), S_i \rangle = 0.$$

$$c_1(\mathfrak{s}) = 2w + (2k+1)e, \quad w \in (-E_8), \quad e^2 = -1, \quad c_1(\mathfrak{s})^2 = 4w^2 - (2k+1)^2 \leq -1 < 7.$$

Hence X^* has no Seiberg–Witten basic class. $\square$

The same obstructions yield minimal-genus, geography, triangulation, contact, surface-singularity, and Dehn-surgery consequences [31, 12, 23, 22, 29, 17].

$$(X, \omega), \quad [\Sigma] = [\Sigma_\omega] \implies g(\Sigma) \geq g(\Sigma_\omega),$$

$$X \text{ spin and indefinite} \implies b_2(X) \geq \frac{5}{4}|\sigma(X)| + 2,$$

$$S_0^3(K) \cong_{\text{or}} S^1 \times S^2 \implies K = U, \quad r \neq 0, \quad S_r^3(K) \cong_{\text{or}} S_r^3(U) \implies K = U.$$

1.2. Torus, Knot, and Rim Surgery. Write $E(M) = X \setminus \text{int}(\nu M)$, choose peripheral bases (μ_K, λ_K) and (μ_T, α, β) , and reverse boundary orientation in every gluing [21, 27, 7, 9].

Definition 1.2.1 (Torus surgery, [21]). For a framed square-zero torus $T \subseteq X^4$, a primitive triple (p, q, r) , and

$$g_*[\partial D^2] = p\mu_T + q\alpha + r\beta, \quad \gcd(p, q, r) = 1,$$

set

$$X_{p,q,r}(T) := E(T) \cup_g (D^2 \times T^2).$$

The slopes (p, q, r) and $(-p, -q, -r)$ determine the same unoriented filling.

Proposition 1.2.2 (Slope and product formulae, [21]). Write $X_g(T) := E(T) \cup_g (D^2 \times T^2)$. For gluings g, g' and $T = K \times S^1 \subseteq S^3 \times S^1$,

$$g_*[\partial D^2] = g'_*[\partial D^2] \implies X_g(T) \cong_{\text{sm}} X_{g'}(T),$$

$$(S^3 \times S^1)_{p,q,0}(K \times S^1) \cong S_{p/q}^3(K) \times S^1.$$

Proof. The common meridian image reduces the first claim to an extending boundary map.

$$h = g'^{-1} \circ g, \quad g' \circ h = g, \quad h_* = \begin{pmatrix} 1 & m & n \\ 0 & a & b \\ 0 & c & d \end{pmatrix}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}),$$

$$H(re^{i\theta}, t_1, t_2) = \left(re^{i(\theta + mt_1 + nt_2)}, at_1 + bt_2, ct_1 + dt_2 \right), \quad H|_{T^3} = h,$$

$$E(T) \cup_g (D^2 \times T^2) \xrightarrow{\text{id} \cup H} E(T) \cup_{g'} (D^2 \times T^2).$$

$$(S^3 \times S^1)_{p,q,0}(K \times S^1) = (E(K) \cup_{p\mu_K + q\lambda_K} (D^2 \times S^1)) \times S^1 = S_{p/q}^3(K) \times S^1.$$

Thus the slope and product formulae follow. □

Every logarithmic transform preserves Euler characteristic and signature:

$$\chi(D^2 \times T^2) = \sigma(D^2 \times T^2) = 0 \implies (\chi, \sigma)(X_{p,q,r}(T)) = (\chi, \sigma)(X).$$

Theorem 1.2.3 (Baykur–Sunukjian, [1]). For closed oriented smooth four-manifolds, with one endpoint simply connected,

$$\left. \begin{array}{l} \chi(X) = \chi(X'), \quad \sigma(X) = \sigma(X'), \\ \pi_1(X) = 1 \text{ or } \pi_1(X') = 1 \end{array} \right\} \xLeftrightarrow{1.2.1} X \xrightarrow{\text{finitely many log transforms}} X'.$$

Open Problem. Realize the sequence in Theorem 1.2.3 on one disjoint torus link [1, 21].

Write $Tw_{n,-m}$ for the generalized twist knot obtained from the n -twist pattern J_n by the displayed $1/m$ -Rolfen twist.

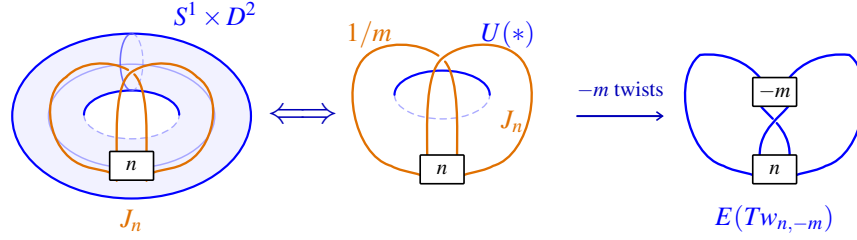


FIGURE 2. A Rolfen twist gives $(S^1 \times D^2)_{1/m}(J_n) \cong E(Tw_{n,-m})$.

Proposition 1.2.4 (Twist-knot torus surgery, [14, 21]). *For $T_n = J_n \times S^1 \subseteq (S^1 \times D^2) \times S^1$,*

$$((S^1 \times D^2) \times S^1)_{1,m,0}(T_n) \cong ((S^1 \times D^2)_{1/m}(J_n)) \times S^1 \cong E(Tw_{n,-m}) \times S^1.$$

Proof. The zero third coefficient and Figure 2 give the complete replacement.

$$((S^1 \times D^2) \times S^1)_{1,m,0}(J_n \times S^1) \xrightarrow{r=0} (S^1 \times D^2)_{1/m}(J_n) \times S^1 \xrightarrow{\text{Rolfen twist}} E(Tw_{n,-m}) \times S^1.$$

Thus one torus realizes the twist-knot replacement. $\square$

Proposition 1.2.5 (Unknotted-torus invariance, [21]). *If $T \subseteq X$ bounds a standard solid torus in a four-ball and has the induced product framing, then*

$$X_{1,q,r}(T) \cong_{\text{sm}} X, \quad q, r \in \mathbb{Z}.$$

Proof. Standard Kirby slides are supported in the four-ball and fix its boundary.

$$(B^4, T)_{1,q,r} \xrightarrow{\text{two handle slides}} (B^4, T)_{1,0,0} \xrightarrow{\cong} B^4$$

$$X_{1,q,r}(T) = (X \setminus \text{int}(B^4)) \cup_{S^3} (B^4, T)_{1,q,r} \cong (X \setminus \text{int}(B^4)) \cup_{S^3} B^4 = X.$$

Thus the ambient manifold is unchanged. $\square$

Example 1.2.6 (Dolgachev surfaces, [6, 21]). For a regular elliptic fiber $T_f \subseteq E(1) = \mathbb{CP}^2 \# 9\overline{\mathbb{CP}}^2$,

$$E(1) \xrightarrow{p\text{-log on } T_f} E(1)_p \xrightarrow{q\text{-log on } T_f} E(1)_{p,q}, \quad \gcd(p, q) = 1,$$

$$\mathcal{Q}_{E(1)_{p,q}} \cong \mathcal{Q}_{E(1)}, \quad E(1)_{p,q} \cong_{\text{homeo}} E(1), \quad E(1)_{p,q} \not\cong_{\text{sm}} E(1) \quad (p, q > 1).$$

Theorem 1.2.7 (Morgan–Mrowka–Szabó, [27]). *Let $T \subseteq X$ be parametrized and square-zero, assume the relevant fillings have $b^+ > 1$, and fix $\mathfrak{t} \in \text{Spin}^c(E(T))$ with $c_1(\mathfrak{t})|_{T^3} = 0$. Set*

$$\mathcal{S}_{a,b,c}(\mathfrak{t}) := \sum_{\substack{\mathfrak{s} \in \text{Spin}^c(X_{a,b,c}(T)) \\ \mathfrak{s}|_{E(T)} = \mathfrak{t}}} \text{SW}_{X_{a,b,c}(T)}(\mathfrak{s}).$$

With compatible homology orientations,

$$\mathcal{S}_{p,q,r}(\mathfrak{t}) = p\mathcal{S}_{1,0,0}(\mathfrak{t}) + q\mathcal{S}_{0,1,0}(\mathfrak{t}) + r\mathcal{S}_{0,0,1}(\mathfrak{t}).$$

The chambered formula holds for $b^+ = 1$ after choosing compatible chambers.

For $b^+(X) > 1$, write

$$\mathcal{SW}_X := \sum_{\mathfrak{s} \in \text{Spin}^c(X)} \text{SW}_X(\mathfrak{s}) e^{c_1(\mathfrak{s})} \in \mathbb{Z}[H^2(X; \mathbb{Z})].$$

Example 1.2.8 (Borromean calculation, [28, 27, 25]). Let $Y = \#_2(S^2 \times S^1)$, let $B \subseteq Y$ be the remaining Borromean component after zero surgery on the other two, and set

$$X = Y \times S^1, \quad T = B \times S^1.$$

Let $N_{n,1} \rightarrow T^2$ be the oriented circle bundle of Euler class n , and let τ_Y denote Turaev torsion. Then

$$\begin{aligned} X_{1,0,0}(T) &= X, & X_{0,1,0}(T) &= T^4, & X_{n,1,0}(T) &= N_{n,1} \times S^1, & e(N_{n,1} \rightarrow T^2) &= n, \\ \text{SW}_X &= 0, & \text{SW}_{T^4}(0) &= 1, & \mathcal{S}_{p,q,0} &\xrightarrow{1,2,7} p\mathcal{S}_{1,0,0} + q\mathcal{S}_{0,1,0} = q, \\ q &= \sum_t \text{SW}_{Y_{p/q}(B) \times S^1}(t), & \mathcal{SW}_{Y \times S^1} &\doteq \tau_Y, & g_*[\partial D^2] &\neq \mu_T \implies b_1(X_{p,q,r}(T)) \leq b_1(X). \end{aligned}$$

Here $\doteq$ means equality up to sign and a group-ring unit.

Definition 1.2.9 (Fintushel–Stern knot surgery, [9]). For a parametrized framed square-zero torus $T \subseteq X$, set

$$X(T, K) := E(T) \cup_g (E(K) \times S^1), \quad g_*(\lambda_K) = \mu_T, \quad g_*(\mu_K) = \alpha, \quad g_*[S^1] = \beta.$$

The peripheral parametrization is part of the data [9, 21].

Theorem 1.2.10 (Knot-complement theorem, [13]).

$$E(K) \cong_{\text{or}} E(J) \implies K = J.$$

Example 1.2.11 (Product knot surgery, [9]). For $T = \text{pt} \times T^2 \subseteq S^2 \times T^2$,

$$(S^2 \times T^2)(T, K) = ((D^2 \times S^1) \cup_{\partial D^2 \mapsto \lambda_K} E(K)) \times S^1 = S_0^3(K) \times S^1.$$

Proposition 1.2.12 (Topology of knot surgery, [10, 9]). For every knot K ,

$$H_*(X(T, K); \mathbb{Z}) \cong H_*(X; \mathbb{Z}), \quad Q_{X(T, K)} \cong Q_X.$$

If $\pi_1(E(T)) = 1$, then

$$\pi_1(X) = \pi_1(X(T, K)) = 1, \quad X(T, K) \cong_{\text{homeo}} X.$$

Proof. Choose a degree-one peripheral-preserving map of pairs $\phi : E(K) \rightarrow E(U)$. Its homology isomorphism identifies the Mayer–Vietoris sequences.

$$\begin{array}{ccccccc} H_k(T^3) & \longrightarrow & H_k(E(T)) \oplus H_k(E(K) \times S^1) & \longrightarrow & H_k(X(T, K)) & \longrightarrow & H_{k-1}(T^3) \\ \parallel & & \cong \downarrow (\text{id} \oplus \phi \times \text{id})_* & & \downarrow & & \parallel \\ H_k(T^3) & \longrightarrow & H_k(E(T)) \oplus H_k(E(U) \times S^1) & \longrightarrow & H_k(X) & \longrightarrow & H_{k-1}(T^3). \end{array}$$

$$H_*(X(T, K); \mathbb{Z}) \cong H_*(X; \mathbb{Z}), \quad F := \text{id}_{E(T)} \cup (\phi \times \text{id}).$$

$$F_*[X(T, K)] = [X], \quad F^* : H^2(X) \xrightarrow{\cong} H^2(X(T, K)) \implies Q_{X(T, K)}(F^*a, F^*b) = Q_X(a, b),$$

$$\langle\langle \mu_K \rangle\rangle = \pi_1(E(K)), \quad g_*(\mu_K) = \alpha = 1, \quad g_*[S^1] = \beta = 1,$$

$$\pi_1(E(T)) = 1 \implies \pi_1(X) = \pi_1(X(T, K)) = 1 \xrightarrow{1.1.2} X(T, K) \cong_{\text{homeo}} X.$$

Thus homology, form, fundamental group, and homeomorphism type are fixed. $\square$

Theorem 1.2.13 (Fintushel–Stern knot-surgery formula, [9]). Let X be closed and simply connected with $b^+(X) > 1$, and let $T \subseteq X$ be a homologically essential c -embedded square-zero torus, meaning a regular fiber in a cusp neighborhood, with $\pi_1(E(T)) = 1$. For the normalized symmetric Alexander polynomial,

$$(KS) \quad \mathcal{SW}_{X(T, K)} = \mathcal{SW}_X \Delta_K(t), \quad t = e^{2\text{PD}([T])}.$$

Theorem 1.2.14 (Elliptic-fiber knot surgery, [9]). *Every regular fiber $T \subseteq E(2) = K3 \rightarrow S^2$ is π_1 -null:*

$$E(2) \setminus \nu T \longrightarrow D^2, \quad \varphi_{\partial D^2} = (tat_b)^{12}, \quad \pi_1(E(2) \setminus \nu T) = \pi_1(T^2) / \langle \langle a, b \rangle \rangle = 1.$$

Consequently,

$$K3(T, K) \cong_{\text{homeo}} K3, \quad \Delta_K \neq 1 \implies K3(T, K) \not\cong_{\text{sm}} K3.$$

Proof. The two vanishing cycles normally generate the fiber group.

$$\pi_1(T^2) = \langle a, b \mid [a, b] \rangle \xrightarrow{a=b=1} \pi_1(E(2) \setminus \nu T) = 1 \xrightarrow{1.2.12} K3(T, K) \cong_{\text{homeo}} K3.$$

$$\mathcal{SW}_{K3} = 1 \xrightarrow{1.2.13} \mathcal{SW}_{K3(T, K)} = \Delta_K(e^{2\text{PD}([T])}) \neq 1 \xrightarrow{1.1.8} K3(T, K) \not\cong_{\text{sm}} K3.$$

Thus $\Delta_K \neq 1$ changes only the smooth structure. $\square$

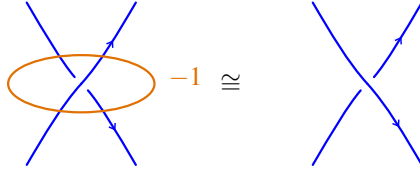


FIGURE 3. (-1) -surgery changes the enclosed crossing.

Proposition 1.2.15 (Crossing-circle realization, [9]). *If disjoint crossing circles $c_1, \dots, c_N \subseteq E(U)$, with coefficients ε_i , turn U into K , then*

$$\begin{aligned} c_i \times S^1 &\subseteq E(U) \times S^1 \cong D^2 \times T^2, & [c_i \times S^1] &= 0, \\ E(U) \times S^1 &\xrightarrow{\{\varepsilon_i\text{-log transforms on } c_i \times S^1\}} E(K) \times S^1, \\ X &= X_0 \xrightarrow{T_1} X_1 \xrightarrow{T_2} \dots \xrightarrow{T_N} X_N = X(T, K). \end{aligned}$$

Proof. By Figure 3, crossing changes become commuting product surgeries.

$$\begin{array}{ccc} (E(U), U) & \xrightarrow{\varepsilon_i\text{-surgery on } c_i} & (E(K), K) \\ \uparrow /S^1 & & \uparrow /S^1 \\ (E(U) \times S^1, U \times S^1) & \xrightarrow{\varepsilon_i\text{-log on } c_i \times S^1} & (E(K) \times S^1, K \times S^1) \end{array}$$

$$[c_i \times S^1] = [c_i] \times [S^1] = 0, \quad (c_i \times S^1) \cap (c_j \times S^1) = \emptyset \quad (i \neq j).$$

Thus knot surgery is a sequence of log transforms on disjoint nullhomologous tori. $\square$

Fact 1.2.16 (Open knot-surgery problems, [9, 21, 33]). *Write $X(T, K; g)$ when the peripheral gluing is g . For simply connected X and $\pi_1(E(T)) = 1$, the assertions*

$$X(T, K; g) \cong_{\text{sm}} X(T, K; g'), \quad \Delta_K = 1 \implies X(T, K) \cong_{\text{sm}} X$$

remain open; the second fails for $\pi_1(X) = \mathbb{Z}$ [20]. For a cusp neighborhood C , it is also open whether unrelated K, J satisfy

$$C(T, K) \cong_{\text{sm}} C(T, J).$$

Proposition 1.2.17 (Single-torus realization and exotica, [14]). *Let $S_n \subseteq \nu T$ be the image of $J_n \times S^1$ under $\nu T \cong D^2 \times T^2$. Then*

$$X_{1,m,0}(S_n) \cong X(T, Tw_{n,-m}).$$

Consequently one torus surgery can preserve homeomorphism type and change smooth structure.

Proof. The marked local replacement is exactly the knot-surgery gluing.

$$\begin{aligned} \nu T_{1,m,0}(S_n) &\xrightarrow{2} E(Tw_{n,-m}) \times S^1, & (\mu_{Tw}, \lambda_{Tw}, s) &\mapsto (\alpha, \mu_T, \beta), \\ X_{1,m,0}(S_n) &\xleftrightarrow{1.2.9} X(T, Tw_{n,-m}), \\ \Delta_{Tw_{n,-m}} \neq 1 &\xrightarrow{1.2.14} K3_{1,m,0}(S_n) \begin{cases} \cong_{\text{homeo}} K3, \\ \not\cong_{\text{sm}} K3. \end{cases} \end{aligned}$$

Thus a single torus detects exotic smooth structure. $\square$

Theorem 1.2.18 (Topological surface classification, [34, 3]). *Let $F_0, F_1 \subseteq X$ be locally flat closed surfaces of equal genus and homology class in a simply connected closed four-manifold. If*

$$\pi_1(X \setminus F_i) \cong \mathbb{Z}, \quad b_2(X) \geq |\sigma(X)| + 6,$$

then $F_0 \simeq_{\text{top}} F_1$. More generally, let F_0, F_1 be homologous oriented surfaces of equal genus with infinite-cyclic exteriors, and let $\lambda_{E(F_i)}$ denote their equivariant intersection forms. A boundary-compatible isometry satisfies

$$\lambda_{E(F_0)} \cong_{\partial} \lambda_{E(F_1)} \text{ over } \mathbb{Z}[t^{\pm 1}] \implies (X, F_0) \cong_{\text{top}} (X, F_1).$$

Theorem 1.2.19 (Hoffman–Sunukjian, [14]). *Let $T \subseteq K3$ be an elliptic fiber, let $S = S_2 \subseteq \nu T$ be the twist torus, and let S' be the unknotted torus. Then*

$$\pi_1(K3 \setminus S') \cong \pi_1(K3 \setminus S) \cong \mathbb{Z},$$

and S, S' are topologically unknotted nullhomologous tori which are not smoothly equivalent.

Proof. The pattern quotient computes both complement groups [14].

$$\begin{aligned} P_n &:= (S^1 \times D^2) \setminus \nu J_n, & K3 \setminus \nu S_n &= (K3 \setminus \nu T) \cup_{T^3} (P_n \times S^1), \\ \pi_1(K3 \setminus \nu S_n) &= \frac{\pi_1(P_n) \times \langle s \rangle}{\langle \langle \mu_{\partial}, \lambda_{\partial}, s \rangle \rangle} \xrightarrow{2} \langle \mu_{J_n} \rangle \cong \mathbb{Z}. \end{aligned}$$

For $\Delta_{Tw_{2,-m}} \neq 1$, the two local surgeries distinguish the smooth embeddings.

$$\begin{aligned} K3_{1,m,0}(S) &\xrightarrow{1.2.17} K3(T, Tw_{2,-m}) \xrightarrow{1.2.14} K3_{1,m,0}(S) \not\cong_{\text{sm}} K3, \\ K3_{1,m,0}(S') &\xrightarrow{1.2.5} K3 \implies S \not\cong_{\text{sm}} S', \\ [S] = [S'] &= 0, \quad g(S) = g(S') = 1, \quad \pi_1(K3 \setminus S) = \pi_1(K3 \setminus S') = \mathbb{Z}, \\ b_2(K3) &= 22 = |\sigma(K3)| + 6 \xrightarrow{1.2.18} S \simeq_{\text{top}} S'. \end{aligned}$$

Thus S, S' form an exotic topologically unknotted pair. $\square$

Definition 1.2.20 (Rim torus and rim surgery, [7]). For $\Sigma \subseteq X$ and a homologically essential circle $\sigma \subseteq \Sigma$, set

$$T_{\sigma} := \partial(\nu \Sigma|_{\sigma}), \quad \Sigma_K := \Sigma(T_{\sigma}, K).$$

Lemma 1.2.21 (Bounding-torus knot surgery, [7]). *If $T = \partial(S^1 \times D^2) \subseteq X$ has its induced parametrization, then*

$$X(T, K) \cong_{\text{sm}} X.$$

Proof. The product parametrization reduces the replacement to meridional filling.

$$(S^1 \times D^3)(T, K) \xlongequal{\quad} S^1 \times ((D^3 \setminus \nu c) \cup_{\lambda_K \mapsto \mu_c, \mu_K \mapsto \lambda_c} E(K)) \xrightarrow{\text{meridional filling}} S^1 \times D^3.$$

Thus the local pair, and hence X , is unchanged. $\square$

Theorem 1.2.22 (Fintushel–Stern rim surgery, [7, 8, 24]). *Let X be closed symplectic and let $\Sigma \subseteq X$ be a positive-genus symplectic surface with*

$$\pi_1(E(\Sigma)) = 1, \quad \Sigma^2 \geq 0, \quad [\sigma] \neq 0 \in H_1(\Sigma).$$

If $\Delta_K \neq 1$, then topological surgery and relative Seiberg–Witten theory give [2, 7, 8]

$$\Sigma(T_\sigma, K) \simeq_{\text{top}} \Sigma, \quad \Sigma(T_\sigma, K) \not\simeq_{\text{sm}} \Sigma.$$

The same conclusion holds under Mark’s hypotheses

$$\Sigma^2 \geq 2 - 2g(\Sigma), \quad \Delta_K(t) \not\equiv 1 \pmod{2}.$$

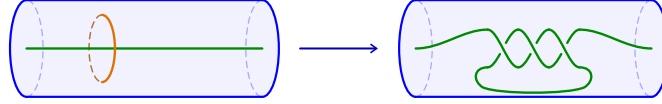


FIGURE 4. Knot surgery replaces the core arc by a knotted arc.

Proposition 1.2.23 (Kim’s tangle model, [16]). *For*

$$\begin{aligned} (\nu A, A) &\cong (D^2 \times I \times S^1, \{0\} \times I \times S^1), & T_\mu &= \mu \times S^1, \\ \{0\} \times I \times S^1 &\xrightarrow{\text{knot surgery on } T_\mu} K^\circ \times S^1. \end{aligned}$$

The ambient manifold is fixed and $\Sigma_K = \Sigma(T_\mu, K)$.

Proof. The S^1 -equivariant replacement is displayed in Figure 4.

$$\begin{aligned} (D^2 \times I \times S^1)(T_\mu, K) &= ((D^2 \times I \setminus \nu\mu) \cup E(K)) \times S^1 \cong (D^2 \times I) \times S^1, \\ \{0\} \times I \times S^1 &\longmapsto K^\circ \times S^1 \implies \Sigma \longmapsto \Sigma(T_\mu, K) = \Sigma_K. \end{aligned}$$

Thus Kim’s tangle and rim-surgery models agree. □

Proposition 1.2.24 (Surface-neighborhood regluing, [21]). *Write*

$$D(\nu\Sigma) = \nu\Sigma, \quad S(\nu\Sigma) = \partial\nu\Sigma.$$

Nonextension can occur only in the following regimes:

Σ	$S(\nu\Sigma)$	potential obstruction
$S^2, \Sigma^2 = 0$	$S^2 \times S^1$	$\pi_1(\text{SO}(3)) = \mathbb{Z}/2$
$T^2, \Sigma^2 = 0$	T^3	$A \in \text{SL}_3(\mathbb{Z}), A\langle\mu\rangle \neq \langle\mu\rangle$
$\mathbb{RP}^2, e = \pm 2$	$S(\nu\mathbb{RP}^2)$	Price
Klein bottle, $e = 0$	$S(\nu\Sigma)$	Klein-bottle

These are the Gluck, logarithmic-transform, Price-twist, and Klein-bottle regimes.

Proof. Base maps and fiber rotations extend radially over the disk bundle.

$$\Phi_\partial(x, e^{i\theta}) = (f(x), e^{i(\theta + \rho(x))}), \quad \Phi_D(x, re^{i\theta}) = (f(x), re^{i(\theta + \rho(x))}),$$

$$\begin{array}{ccc} S(\nu\Sigma) & \xrightarrow{\Phi_\partial} & S(\nu\Sigma) \\ \downarrow & & \downarrow \\ D(\nu\Sigma) & \xrightarrow{\Phi_D} & D(\nu\Sigma). \end{array}$$

$$\ker(H_1(T^3) \rightarrow H_1(T^2 \times D^2)) = \langle\mu\rangle, \quad A \text{ extends over } T^2 \times D^2 \iff A\langle\mu\rangle = \langle\mu\rangle.$$

Thus the radial classes extend, leaving the four possible regimes displayed above. □

Open Problem. Construct a higher-genus surgery with a relative Seiberg–Witten formula which preserves simple connectivity for π_1 -null surfaces [21].

2. CORKS AND FLOER-THEORETIC EXOTICA

Fix a closed, connected, oriented three-manifold Y , put $\mathbb{F} = \mathbb{Z}/2$, and require every boundary map to preserve orientation.

2.1. Corks and Exotica with Boundary.

Theorem 2.1.1 (Cork theorem, [21]). *Let X, X' be compact, simply connected, homeomorphic, and nondiffeomorphic, with the boundary identification fixed in the relative case. Then*

$$\exists C \subseteq X, \quad C \text{ contractible}, \quad \tau: \partial C \longrightarrow \partial C, \quad \tau \in \text{Diff}^+(\partial C), \quad \tau^2 = \text{id}_{\partial C}.$$

$$X' \cong_{\text{sm}} (X \setminus \text{int}(C)) \cup_{\tau} C.$$

Definition 2.1.2 (Involutive cork, [21]). A τ -cork is a compact contractible four-manifold C and an orientation-preserving involution $\tau: \partial C \rightarrow \partial C$ such that

$$\tau \text{ extends homeomorphically over } C, \quad \tau \text{ does not extend diffeomorphically over } C.$$

For $C \subseteq X$, set

$$X^{\tau} := (X \setminus \text{int}(C)) \cup_{\tau} C.$$

Theorem 2.1.3 (Topological extension, [10, 11, 21]). *If C is compact and contractible, then*

$$\forall \tau \in \text{Homeo}^+(\partial C), \quad \exists \bar{\tau} \in \text{Homeo}^+(C), \quad \bar{\tau}|_{\partial C} = \tau,$$

hence every cork twist satisfies

$$X^{\tau} \cong_{\text{homeo}} X.$$

For a slice disk $D_K \subseteq B^4$ of $K \subseteq S^3$, write

$$E(K) := S^3 \setminus \text{int}(\nu K), \quad E(D_K) := B^4 \setminus \text{int}(\nu D_K), \quad \partial E(D_K) = S_0^3(K).$$

Proposition 2.1.4 (Slice-disk contractible, [21]). *Let C_{D_K} be obtained from $E(D_K)$ by attaching a (-1) -framed 2-handle along μ_K . Then*

$$\partial C_{D_K} = S_{+1}^3(K), \quad C_{D_K} \simeq \text{pt},$$

and C_{D_K} is the (-1) -framed surgery on D_K .

Proof. The handle attachment simultaneously changes the boundary framing and kills the normal generator.

$$\begin{aligned} E(D_K) &\xrightarrow{h_{\mu_K}^2(-1)} C_{D_K} \xrightarrow{\partial} S_{0,-1}^3(K \cup \mu_K) \xrightarrow{\text{blow down}} S_{+1}^3(K) \\ H_2(C_{D_K}, E(D_K)) &\cong \mathbb{Z} \xrightarrow{[\text{core}] \mapsto [\mu_K]} H_1(E(D_K)) \cong \mathbb{Z}, \quad H_i(C_{D_K}) = 0 \quad (i > 0), \\ \pi_1(C_{D_K}) &= \pi_1(E(D_K)) / \langle\langle \mu_K \rangle\rangle = 1 \xrightarrow{\text{Hurewicz-Whitehead}} C_{D_K} \simeq \text{pt}, \\ D_K \subset B^4 &\xrightarrow{-1 \text{ boundary twist}} D_K(-1) \subset C_{D_K} \xlongequal{\quad} (-1)\text{-framed disk surgery} \end{aligned}$$

Thus C_{D_K} has the asserted boundary, homotopy type, and disk-surgery description. $\square$

Proposition 2.1.5 (Strong inversion after surgery, [21]). *A strong inversion $\iota: (S^3, K) \rightarrow (S^3, K)$ induces*

$$\hat{\iota}: S_{+1}^3(K) = \partial C_{D_K} \longrightarrow S_{+1}^3(K).$$

Proof. Invariant peripheral coordinates fix the unoriented $(+1)$ -slope.

$$\begin{aligned}
 \iota(z, e^{i\theta}) &= (\bar{z}, e^{-i\theta}), & \iota_*(\mu_K) &= -\mu_K, & \iota_*(\lambda_K) &= -\lambda_K, \\
 \iota_*(\mu_K + \lambda_K) &= -(\mu_K + \lambda_K), & \{\pm(\mu_K + \lambda_K)\} &\text{ is fixed,} \\
 \partial E(K) &\xrightarrow{\iota|_{\partial E(K)}} \partial E(K) \\
 \downarrow & & \downarrow \\
 E(K) \cup_{\mu_K + \lambda_K} (D^2 \times S^1) &\xrightarrow{\hat{\iota}} E(K) \cup_{\mu_K + \lambda_K} (D^2 \times S^1)
 \end{aligned}$$

Hence ι extends across the filling solid torus and induces $\hat{\iota}$. □

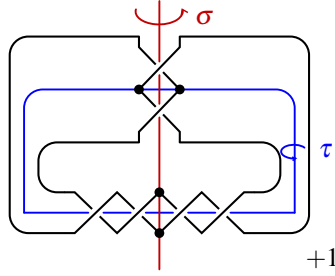


FIGURE 5. The strong inversions of the stevedore knot 6_1 [21].

Theorem 2.1.6 (Heegaard Floer functoriality, [21]). *For $(Y, \mathfrak{s})$,*

$$\mathrm{HF}^-(Y, \mathfrak{s}) \text{ is an } \mathbb{F}[U]\text{-module,} \quad \mathrm{HF}^-(S^3) \cong \mathbb{F}[U].$$

A connected Spin^c -cobordism $(W, \mathfrak{t}) : (Y_0, \mathfrak{s}_0) \rightarrow (Y_1, \mathfrak{s}_1)$ induces

$$F_{W, \mathfrak{t}}^- : \mathrm{HF}^-(Y_0, \mathfrak{s}_0) \longrightarrow \mathrm{HF}^-(Y_1, \mathfrak{s}_1),$$

and for composable cobordisms,

$$F_{W', \mathfrak{t}'}^- \circ F_{W, \mathfrak{t}}^- = \sum_{\substack{\mathfrak{u}|_W = \mathfrak{t} \\ \mathfrak{u}|_{W'} = \mathfrak{t}'}} F_{W \cup W', \mathfrak{u}}^-.$$

If an orientation-preserving diffeomorphism $\Phi : W_0 \rightarrow W_1$ restricts to ϕ_0, ϕ_1 , then

$$\begin{array}{ccc}
 \mathrm{HF}^-(Y_0, \mathfrak{s}_0) & \xrightarrow{(\phi_0)_*} & \mathrm{HF}^-(Y'_0, (\phi_0)_* \mathfrak{s}_0) \\
 \downarrow F_{W_0, \mathfrak{t}}^- & & \downarrow F_{W_1, \Phi_* \mathfrak{t}}^- \\
 \mathrm{HF}^-(Y_1, \mathfrak{s}_1) & \xrightarrow{(\phi_1)_*} & \mathrm{HF}^-(Y'_1, (\phi_1)_* \mathfrak{s}_1).
 \end{array}$$

Definition 2.1.7 (Relative invariant, [21]). For compact, connected, oriented W with connected boundary Y , remove an interior ball and define

$$W^\circ : S^3 \rightarrow Y, \quad r(W, \mathfrak{t}) := F_{W^\circ, \mathfrak{t}}^-(1) \in \mathrm{HF}^-(Y, \mathfrak{t}|_Y).$$

Proposition 2.1.8 (Relative naturality, [21]). *If $\phi : Y \rightarrow Y$ extends to an orientation-preserving diffeomorphism $\Phi : W \rightarrow W$, then*

$$\phi_* r(W, \mathfrak{t}) = r(W, \Phi_* \mathfrak{t}).$$

Proof. Fix the puncturing ball and apply naturality to W° .

$$\begin{array}{ccc}
 \mathrm{HF}^-(S^3) & \xrightarrow{\mathrm{id}} & \mathrm{HF}^-(S^3) \\
 F_{W^\circ, \mathfrak{t}}^- \downarrow & & \downarrow F_{W^\circ, \Phi_* \mathfrak{t}}^- \\
 \mathrm{HF}^-(Y, \mathfrak{t}|_Y) & \xrightarrow{\phi_*} & \mathrm{HF}^-(Y, (\Phi_* \mathfrak{t})|_Y)
 \end{array}$$

$$1 \mapsto F_{W^\circ, \mathfrak{t}}^-(1) = r(W, \mathfrak{t}), \quad 1 \mapsto F_{W^\circ, \Phi_* \mathfrak{t}}^-(1) = r(W, \Phi_* \mathfrak{t}),$$

$$\phi_* F_{W^\circ, \mathfrak{t}}^-(1) = F_{W^\circ, \Phi_* \mathfrak{t}}^-(1).$$

Thus evaluation at $1 \in \mathbb{F}[U]$ gives the asserted identity. $\square$

For $Y = S_1^3(6_1)$, the Dai–Hedden–Mallick computation is [4]

$$\begin{aligned}
 t_1 &:= x + y, & t_2 &:= x + z, & \mathrm{HF}^-(Y) &\cong \mathbb{F}[U] \cdot y \oplus \mathbb{F} \cdot t_1 \oplus \mathbb{F} \cdot t_2, \\
 Ut_1 = Ut_2 &= 0, & \tau_*(y) &= y + t_2, & \tau_*(t_1) &= t_1, & \tau_*(t_2) &= t_2.
 \end{aligned}$$

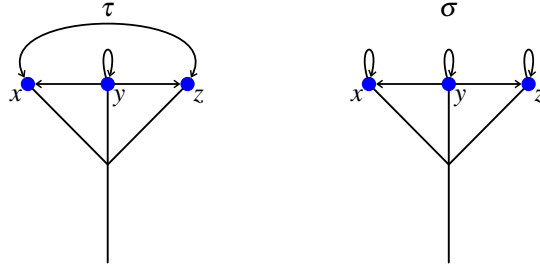


FIGURE 6. The U -action and the involutions τ, σ on $\mathrm{HF}^-(S_1^3(6_1))$ [21].

Theorem 2.1.9 (Contractible relative invariant, [21]). *For every compact contractible four-manifold C , $r(C)$ generates an $\mathbb{F}[U]$ -summand of $\mathrm{HF}^-(\partial C)$.*

Proof. Localization and the absolute grading detect the primitive free-tower coefficient.

$$\begin{array}{ccc}
 \mathrm{HF}^-(S^3) & \xrightarrow{F_{C^\circ}^-} & \mathrm{HF}^-(Y) \\
 \downarrow & & \downarrow \\
 \mathrm{HF}^\infty(S^3) & \xrightarrow{F_{C^\circ}^\infty} & \mathrm{HF}^\infty(Y)
 \end{array}
 \quad (Y = \partial C).$$

$$H_*(C^\circ, S^3) = 0, \quad F_{C^\circ}^\infty \text{ is a degree-zero isomorphism,} \quad d(Y) = d(S^3) = 0,$$

$$\mathcal{T}_U(Y) := \{x \in \mathrm{HF}^-(Y) \mid U^N x = 0 \text{ for some } N\},$$

$$\mathrm{HF}^-(Y) \cong \mathbb{F}[U] \cdot y \oplus \mathcal{T}_U(Y), \quad r(C) = F_{C^\circ}^-(1) = y + t, \quad t \in \mathcal{T}_U(Y),$$

$$\mathrm{HF}^-(Y) = \mathbb{F}[U] \cdot (y + t) \oplus \mathcal{T}_U(Y) = \mathbb{F}[U] \cdot r(C) \oplus \mathcal{T}_U(Y).$$

Thus $r(C)$ generates a free $\mathbb{F}[U]$ -summand. $\square$

Proposition 2.1.10 (Ordinary Floer obstruction for τ , [4, 21]). *The involution τ on $S_1^3(6_1)$ extends over no contractible four-manifold bounded by $S_1^3(6_1)$.*

Proof. An extension would fix the relative invariant and contradict its forced free-tower coefficient.

$$\begin{aligned} \tau \text{ extends over } C &\xrightarrow{2.1.8} \tau_* r(C) = r(C), \\ r(C) &\xrightarrow{2.1.9} y + \alpha t_1 + \beta t_2, \quad \alpha, \beta \in \mathbb{F}, \\ \tau_* r(C) &= \tau_*(y) + \alpha \tau_*(t_1) + \beta \tau_*(t_2) = y + \alpha t_1 + (\beta + 1)t_2 = r(C) + t_2, \\ t_2 \neq 0 &\implies \tau_* r(C) \neq r(C) \implies \perp. \end{aligned}$$

Hence τ has no smooth extension over a contractible filling. $\square$

Theorem 2.1.11 (Strong stevedore corks, [4]). *Neither displayed involution*

$$\tau, \sigma: S_1^3(6_1) \longrightarrow S_1^3(6_1)$$

extends over an integral homology four-ball.

Corollary 2.1.12 (Slice-disk corks, [4, 21]). *For every smooth slice disk D of 6_1 , the pair (C_D, τ) is a cork.*

Proof. Contractibility supplies the topological extension, while Floer theory obstructs the smooth one.

$$\begin{aligned} C_D \simeq \text{pt} &\xrightarrow{2.1.3} \tau \text{ extends homeomorphically over } C_D, \\ C_D \simeq \text{pt} &\implies H_*(C_D; \mathbb{Z}) \cong H_*(B^4; \mathbb{Z}) \xrightarrow{2.1.11} \tau \text{ does not extend diffeomorphically over } C_D, \\ (C_D, \tau) &\xleftrightarrow{2.1.2} \text{cork}. \end{aligned}$$

Thus every slice disk of 6_1 produces a cork. $\square$

Theorem 2.1.13 (Akbulut–Ruberman, [21]). *For every cork (C, τ) , there is an invertible homology cobordism V leaving ∂C such that*

$$D = C \cup_{\text{id}} V, \quad D' = C \cup_{\tau} V, \quad D \cong_{\text{homeo}} D', \quad D \not\cong_{\text{sm}} D', \quad D \simeq D' \simeq \text{pt}.$$

Definition 2.1.14 (Surface cork, [21]). An ι -surface cork is a properly embedded disk $D \subseteq B^4$, $K := \partial D$, and a strong inversion $\iota: (S^3, K) \rightarrow (S^3, K)$ such that

$$\pi_1(B^4 \setminus D) \cong \mathbb{Z},$$

ι extends homeomorphically over (B^4, D) , ι does not extend diffeomorphically over (B^4, D) .

Proposition 2.1.15 (Topological uniqueness of $\mathbb{Z}$ -disks, [3, 21]). *Any two $\mathbb{Z}$ -disks for the same knot are topologically ambiently isotopic rel boundary; hence every strong inversion extends homeomorphically over the pair.*

Proof. Extend the inversion over B^4 and apply uniqueness to the image disk.

$$\begin{aligned} \bar{\iota}: B^4 &\rightarrow B^4, \quad \bar{\iota}|_{S^3} = \iota, \quad D' := \bar{\iota}(D), \\ \partial D &= \partial D' = K, \quad \pi_1(B^4 \setminus D) \cong \pi_1(B^4 \setminus D') \cong \mathbb{Z}, \\ D' &\simeq_{\text{isot.top}} D \text{ rel } S^3 \implies \exists H_t \in \text{Homeo}(B^4 \text{ rel } S^3), \quad H_1(D') = D, \\ \Phi &:= H_1 \circ \bar{\iota}, \quad \Phi|_{S^3} = \iota, \quad \Phi(D) = D. \end{aligned}$$

Thus Φ is the required topological extension of the pair. $\square$

Lemma 2.1.16 (Branched cover of a $\mathbb{Z}$ -disk, [21]). *If $D \subseteq B^4$ is a $\mathbb{Z}$ -disk, then*

$$W_D := \Sigma_2(B^4, D) \simeq \text{pt}.$$

Proof. The cyclic cover computation gives simple connectivity and acyclicity.

$$E := B^4 \setminus \nu D, \quad \pi_1(E) = \langle \mu \rangle \cong \mathbb{Z}, \quad \pi_1(\tilde{E}) = \langle \mu^2 \rangle = 2\mathbb{Z},$$

$$W_D = \tilde{E} \cup_{\partial} (D^2 \times D^2), \quad \pi_1(W_D) = \langle \mu^2 \rangle / \langle \langle \mu^2 \rangle \rangle = 1,$$

$$\pi_1(E) = \mathbb{Z} \xrightarrow{[3]} \Delta_{\partial D}(t) = 1 \implies H_1(\partial W_D; \mathbb{Z}) = 0,$$

$$\partial W_D \simeq_H S^3, \quad \pi_1(W_D) = 1, \quad \chi(W_D) = 2\chi(B^4) - \chi(D) = 1 \xrightarrow{\text{PL+UCT}} \tilde{H}_*(W_D; \mathbb{Z}) = 0 \xrightarrow{\text{Whitehead}} W_D \simeq \text{pt.}$$

Thus the double branched cover is contractible. $\square$

Proposition 2.1.17 (Branched-cover cork criterion, [21]). *Let D be a $\mathbb{Z}$ -disk with $\pi_D: W_D \rightarrow B^4$. If a lift τ of a strong inversion ι satisfies*

$$\iota \circ \pi_D|_{\partial W_D} = \pi_D \circ \tau, \quad (W_D, \tau) \text{ is a cork},$$

then (D, ι) is a surface cork.

Proof. A smooth pair extension lifts; compose with the deck involution when necessary to obtain boundary value τ .

$$\begin{array}{ccc} W_D & \xrightarrow{\tilde{\iota}} & W_D \\ \pi_D \downarrow & & \downarrow \pi_D \\ (B^4, D) & \xrightarrow{\iota} & (B^4, D) \end{array} \quad \tilde{\iota}|_{\partial W_D} = \tau.$$

$$\iota \text{ extends smoothly over } (B^4, D) \implies \tau \text{ extends smoothly over } W_D \implies \perp,$$

$$D \text{ is a } \mathbb{Z}\text{-disk} \xrightarrow{2.1.15} \iota \text{ extends homeomorphically over } (B^4, D),$$

$$\left. \begin{array}{l} \iota \text{ extends topologically over } (B^4, D), \\ \iota \text{ does not extend smoothly over } (B^4, D) \end{array} \right\} \xLeftrightarrow{2.1.14} (D, \iota) \text{ is a surface cork.}$$

Thus the branched-cover cork obstruction descends to the surface pair. $\square$

Theorem 2.1.18 (Existence of surface corks, [21]). *There exists an ι -surface cork $(D, \iota) \subseteq B^4$.*

Fact 2.1.19 (Surface-cork generation problem, [21]). *It is open whether every pair of exotic surfaces in a simply connected four-manifold is related by a surface-cork twist.*

Let $K_P = \partial\Gamma$, where Γ is the displayed genus-one Seifert surface, $\gamma \subseteq \Gamma$ is unknotted, nonseparating, and zero-framed, and ι is the vertical strong inversion displayed below.

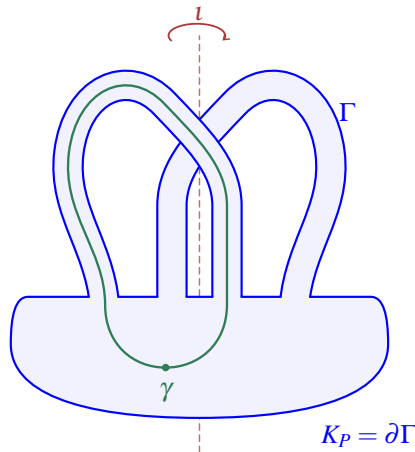


FIGURE 7. The surface Γ , compression curve γ , and strong inversion ι .

Lemma 2.1.20 (The positron disk, [21]). *Compression of Γ along a pushed-in spanning disk for γ gives a smooth $\mathbb{Z}$ -disk $D_P \subseteq B^4$ with $\partial D_P = K_P$.*

Proof. The zero-framed compression and ribbon-complement presentation determine the disk.

$$\Gamma \xrightarrow{\gamma\text{-compression}} D_P \longmapsto \pi_1(B^4 \setminus D_P) = \langle \mu \mid \emptyset \rangle \cong \mathbb{Z}.$$

Hence D_P is a $\mathbb{Z}$ -disk. □

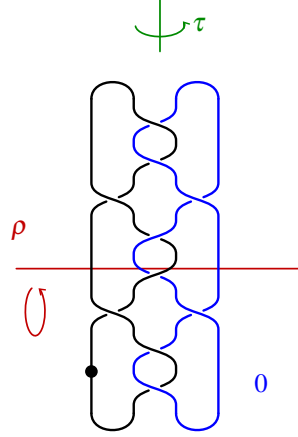


FIGURE 8. The positron P with boundary involution τ and deck involution ρ [21].

Theorem 2.1.21 (The positron cork, [21, 4]). *The displayed handlebody P is contractible and (P, τ) is a cork.*

Proof. The dotted and zero-framed components determine the positron handlebody.

$$\begin{aligned} 0 \longrightarrow C_2(P) = \mathbb{Z} \xrightarrow{\pm 1} C_1(P) = \mathbb{Z} \longrightarrow 0, \quad \pi_1(P) = \langle x \mid x \rangle = 1, \quad \tilde{H}_*(P) = 0. \\ \pi_1(P) = 1, \quad \tilde{H}_*(P) = 0 \xrightarrow{\text{Whitehead}} P \simeq \text{pt}, \quad (\partial P, \tau) \cong (S_{+1}^3(6_1), \tau) \quad [21], \\ (\partial P, \tau) \cong (S_{+1}^3(6_1), \tau) \xrightarrow{2.1.11} \tau \text{ does not extend smoothly over } P. \\ P \simeq \text{pt} \xrightarrow{2.1.3} \tau \text{ extends topologically over } P. \end{aligned}$$

Thus (P, τ) is a cork. □

Definition 2.1.22 (Cut-open branched cover, [21]). For a slice disk $\Delta \subseteq B^4$, choose $M^3 \subseteq B^4$ and put

$$\begin{aligned} \partial M = \Delta \cup_{\partial \Delta} \Gamma, \quad \Gamma \subseteq \partial B^4 = S^3, \quad Q := B^4 \setminus \text{int}(\nu M), \quad \partial Q \supseteq M^+ \sqcup M^-. \\ \Sigma_2(B^4, \Delta) := Q_0 \bigcup_{M_0^+ = M_1^-} Q_1 \bigcup_{M_1^+ = M_0^-}. \end{aligned}$$

For its deck involution ρ_Δ and projection π_Δ ,

$$\text{Fix}(\rho_\Delta) = \pi_\Delta^{-1}(\Delta), \quad \partial \text{Fix}(\rho_\Delta) = \pi_\Delta^{-1}(\partial \Delta).$$

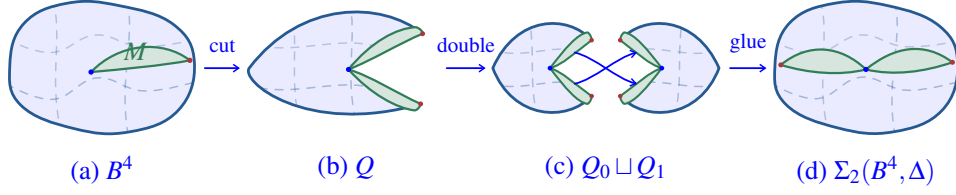


FIGURE 9. Cutting B^4 along M , taking two copies of Q , and crosswise gluing the exposed faces produces $\Sigma_2(B^4, \Delta)$.

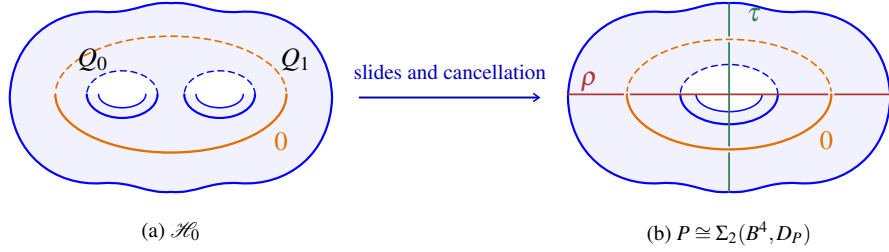


FIGURE 10. Cross-gluing two copies $Q_i \cong S^1 \times B^3$, adding the 0-framed filling curve, and simplifying equivariantly gives $P \cong \Sigma_2(B^4, D_P)$, with deck involution ρ and boundary lift τ .

For D_P , the cut-open model and its handles are

$$M = (\Gamma \times I) \cup_{\gamma} h_{3D}^2 \cong S^1 \times D^2, \quad Q = B^4 \setminus \nu M \xrightarrow{10} \cong B^4 \setminus \nu D^2 \cong S^1 \times B^3, \\ \mathcal{H}(\Sigma_2(B^4, D_P)) \xrightarrow{10} \mathcal{H}_0,$$

where γ is primitive in $\Gamma \times I$.

Proposition 2.1.23 (Positron handle slides, [21]). *The displayed slides transform the branched-cover handlebody into the positron handlebody.*

Proof. The two equivariant slides determine the terminal handlebody.

$$\mathcal{H}(\Sigma_2(B^4, D_P)) = \mathcal{H}_0 \xrightarrow{\text{slide}} \mathcal{H}_1 \xrightarrow{\text{slide}} \mathcal{H}(P), \\ \{h_{\text{fill}}^2, h_{\text{sheet}}^2\} \xrightarrow{\text{clear}} \{h_{\text{clear}}^2, h_{\text{cross}}^2\} \xrightarrow{\text{cross}} \{\dot{K}_P, K_P^0\} \\ f_{\text{new}} = f_1 + f_2 \pm 2\text{lk}(K_1, K_2) = 0.$$

Thus the terminal diagram is P . □

Proposition 2.1.24 (Identification of the positron cover, [21]). *There is a projection*

$$\pi_P: P \cong \Sigma_2(B^4, D_P) \longrightarrow B^4$$

whose deck involution is ρ , and $\tau: \partial P \rightarrow \partial P$ is the boundary lift $\tilde{\iota}$ of ι . Thus

$$\pi_P \circ \rho = \pi_P, \quad \iota \circ \pi_P|_{\partial P} = \pi_P \circ \tau.$$

Proof. The cross-gluing and equivariant slides identify the cover and its actions.

$$\mathcal{H}(\Sigma_2(B^4, D_P)) \xrightarrow{9, 10, 10} \mathcal{H}(P).$$

$$\begin{array}{ccc} \Sigma_2(B^4, D_P) & \xrightarrow{\rho} & \Sigma_2(B^4, D_P) \\ \downarrow \cong & & \downarrow \cong \\ P & \xrightarrow{\rho} & P \\ & \searrow \pi_P & \searrow \pi_P \\ & B^4 & \xlongequal{\quad} B^4 \end{array} \quad \begin{array}{ccc} \partial P & \xrightarrow{\tau} & \partial P \\ \pi_P \downarrow & & \downarrow \pi_P \\ (S^3, K_P) & \xrightarrow{\iota} & (S^3, K_P). \end{array}$$

Hence the identifications are equivariant. □

Theorem 2.1.25 (The positron surface cork, [21]). *The pair (D_P, ι) is a surface cork.*

Proof. The branched-cover criterion applies.

$$D_P \xrightarrow{2.1.20} \pi_1(B^4 \setminus D_P) = \mathbb{Z} \xrightarrow{2.1.16} \Sigma_2(B^4, D_P) \simeq \text{pt},$$

$$(\Sigma_2(B^4, D_P), \tilde{\iota}) \xrightarrow{2.1.24} (P, \tau) \xrightarrow{2.1.21} (\Sigma_2(B^4, D_P), \tilde{\iota}) \text{ is a cork } \xrightarrow{2.1.17} (D_P, \iota) \text{ is a surface cork.}$$

Thus (D_P, ι) is a surface cork. □

Proposition 2.1.26 (Carving criterion, [21]). *Let $\mathcal{H}$ be a handle diagram of B^4 and*

$$K \subseteq \partial \mathcal{H}, \quad K \text{ unknotted and split from the 1-handles}, \quad \pi_1(E(D_K)) \cong \mathbb{Z}.$$

Let D_K be the evident disk, let $\iota: \partial \mathcal{H} \rightarrow \partial \mathcal{H}$ restrict to a strong inversion of K , and let

$$\tau': \partial E(D_K) = S_0^3(K) \longrightarrow S_0^3(K), \quad \tau'': \partial C_{D_K} = S_{+1}^3(K) \longrightarrow S_{+1}^3(K)$$

be the induced involution and its meridional-filling extension. If (C_{D_K}, τ'') is a cork, then (D_K, ι) is a surface cork.

Proof. Carving and meridional filling transfer any smooth pair extension to the cork.

$$(B^4, D_K) \xrightarrow{\setminus \nu D_K} E(D_K) \xrightarrow{+h_{\mu_K, -1}^2} C_{D_K}$$

ι extends smoothly over (B^4, D_K)

$$\Downarrow$$

τ' extends smoothly over $E(D_K)$

$$\Downarrow$$

τ'' extends smoothly over C_{D_K}

$$\pi_1(E(D_K)) = \mathbb{Z} \xrightarrow{2.1.15} \iota \text{ extends topologically over } (B^4, D_K).$$

Hence the cork hypothesis makes (D_K, ι) a surface cork. □

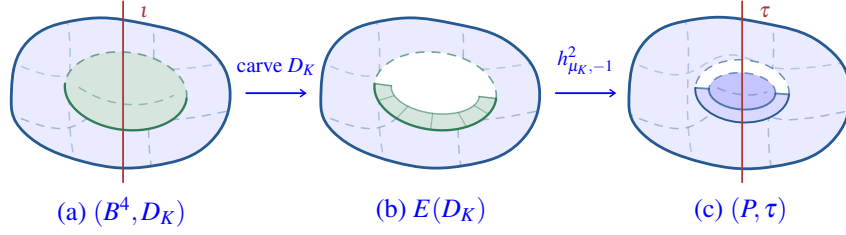


FIGURE 11. Carving the equivariant slice disk D_K and attaching $h^2_{\mu_K, -1}$ yields the positron cork (P, τ) , with $\tau'' = \tau$.

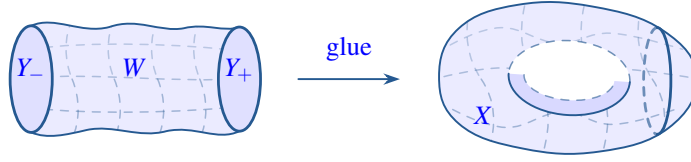


FIGURE 12. The closure $X = W/(Y_- \sim Y_+)$ is obtained by gluing the two boundary copies of Y in the self-cobordism $W: Y \rightarrow Y$.

Proposition 2.1.27 (The carved positron surface cork, [21]). *The displayed disk D_K and vertical strong inversion ι form a surface cork.*

Proof. The diagram verifies both carving hypotheses.

$$\begin{aligned} \pi_1(E(D_K)) &= \langle \mu_K \mid \emptyset \rangle \cong \mathbb{Z}, & (C_{D_K}, \tau'') &\cong (P, \tau) \xrightarrow{2.1.21} (C_{D_K}, \tau'') \text{ is a cork,} \\ (C_{D_K}, \tau'') &\text{ is a cork} \xrightarrow{2.1.26} (D_K, \iota) &\text{ is a surface cork.} \end{aligned}$$

Thus (D_K, ι) is the carved positron surface cork. □

2.2. Closed Floer Theory and Explicit Exotica. For a connected nonseparating $Y \subset X$, write

$$W := X \setminus \text{int}(\nu Y): Y \longrightarrow Y, \quad X = W/(Y_- \sim Y_+), \quad F_W \in \{F_W^{\text{HF}}, F_W^{\text{HM}}\},$$

and let $\Phi_{X,u} \in \mathbb{F}$ denote the closed Heegaard Floer invariant.

Definition 2.2.1 (Reduced Floer homology, [21]). For $n \gg 0$,

$$\begin{aligned} \text{HF}_{\text{red}}(Y, \mathfrak{s}) &:= \ker(U^n: \text{HF}^-(Y, \mathfrak{s}) \rightarrow \text{HF}^-(Y, \mathfrak{s})), & \text{HF}_{\text{red}}(Y, \mathfrak{s}) &= \{x \mid U^N x = 0 \text{ for some } N\}, \\ F_{V, \mathfrak{v}}(\text{HF}_{\text{red}}) &\subseteq \text{HF}_{\text{red}} \end{aligned}$$

for every $\mathbb{F}[U]$ -linear cobordism map.

Theorem 2.2.2 (Closure trace formula, [20, 21]). *Let $(W, \mathfrak{t}): (Y, \mathfrak{s}) \rightarrow (Y, \mathfrak{s})$ close to X , where $b^+(X) > 1$. If*

$$b^+(W) > 0 \quad \text{or} \quad c_1(\mathfrak{s}) \notin \text{Tor} H^2(Y),$$

then

$$\begin{aligned} \sum_{u|_W = \mathfrak{t}} \text{SW}_X(u) &= \text{Lef}_{\mathbb{Z}}^{\text{gr}} \left(F_{W, \mathfrak{t}}^{\text{HM}} \Big|_{\text{HF}_{\text{red}}(Y, \mathfrak{s}; \mathbb{Z})} \right), \\ \sum_{u|_W = \mathfrak{t}} \Phi_{X,u} &= \text{Tr} \left(F_{W, \mathfrak{t}}^{\text{HF}} \Big|_{\text{HF}_{\text{red}}(Y, \mathfrak{s})} \right) \in \mathbb{F}. \end{aligned}$$

Proof. The two gluing laws identify closure counts with Lefschetz numbers.

$$\begin{aligned}
 (W, \mathfrak{t}) : (Y, \mathfrak{s}) &\longrightarrow F_{W, \mathfrak{t}}^{\text{HM}} \xrightarrow{\text{Lef}_{\mathbb{Z}}^{\text{gr}}} \sum_{\mathfrak{u}|_W = \mathfrak{t}} \text{SW}_X(\mathfrak{u}) \\
 (W, \mathfrak{t}) : (Y, \mathfrak{s}) &\longrightarrow F_{W, \mathfrak{t}}^{\text{HF}} \xrightarrow{\text{Lef}_{\mathbb{F}}} \sum_{\mathfrak{u}|_W = \mathfrak{t}} \Phi_{X, \mathfrak{u}}. \\
 -1 = 1 \in \mathbb{F} &\implies \text{Lef}_{\mathbb{F}}(F) = \sum_i (-1)^i \text{Tr}(F_i) = \sum_i \text{Tr}(F_i) = \text{Tr}(F).
 \end{aligned}$$

Thus the monopole and Heegaard Floer trace identities follow. $\square$

Proposition 2.2.3 (Nonseparating three-sphere obstruction, [21]). *If $b^+(X) > 1$ and $S^3 \subset X$ is nonseparating, then*

$$\text{SW}_X \equiv 0.$$

Proof. Cut X along the nonseparating sphere.

$$\begin{aligned}
 W : S^3 &\longrightarrow S^3, \quad b^+(W) = b^+(X) > 1, \quad H^1(S^3; \mathbb{Z}) = H^2(S^3; \mathbb{Z}) = 0, \quad \#\{\mathfrak{u} \mid \mathfrak{u}|_W = \mathfrak{t}\} = 1, \\
 \text{HM}_{\text{red}}(S^3; \mathbb{Z}) &= 0 \xrightarrow{2.2.2} \text{SW}_X(\mathfrak{u}) = \text{Lef}_{\mathbb{Z}}^{\text{gr}}(0) = 0 \quad (\mathfrak{u} \in \text{Spin}^c(X)).
 \end{aligned}$$

Thus every integral Seiberg–Witten invariant vanishes. $\square$

For a framed knot $K \subset Y$, put

$$\widehat{T} := \mathbb{F}[[U]], \quad \mathbf{HF}^-(Y) := \text{HF}^-(Y) \otimes_{\mathbb{F}[U]} \widehat{T}.$$

Functoriality gives the completed surgery triangle

$$\mathbf{HF}^-(Y) \longrightarrow \mathbf{HF}^-(Y_{\lambda}(K)) \longrightarrow \mathbf{HF}^-(Y_{\lambda+\mu}(K)) \longrightarrow \mathbf{HF}^-(Y),$$

whose arrows are the total two-handle maps.

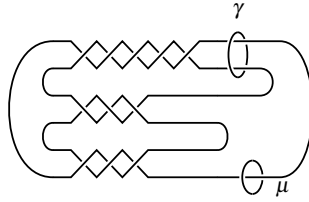


FIGURE 13. The pretzel knot $K_n = P(3, -3, n)$, the curves γ, μ , and the n half-twists.

Proposition 2.2.4 (Reduced surgery map, [20]). *For every displayed knot K_n ,*

$$F_{W_{-1}(\mu)} : \text{HF}_{\text{red}}(S_0^3(K_n)) \xrightarrow{\cong} \text{HF}_{\text{red}}(S_1^3(K_n)).$$

Proof. Put $T = \mathbb{F}[[U]]$ and take every cobordism map total over its Spin^c -structures.

$$\begin{aligned}
 \mathbf{HF}^-(S^3) &= T, \quad \mathbf{HF}^-(S_0^3(K_n)) \cong T^2 \oplus R_0, \quad \mathbf{HF}^-(S_1^3(K_n)) \cong T \oplus R_1, \\
 R_0 &\cong R_1 \cong (T/UT)^2, \quad (T, T^2, T) \otimes_T \mathbb{F}((U)) \cong (\mathbb{F}((U)), \mathbb{F}((U))^2, \mathbb{F}((U))). \\
 r_1 + r_2 &= 2, \quad r_2 + r_3 = 1, \quad r_3 + r_1 = 1 \implies (r_1, r_2, r_3) = (1, 1, 0) \implies F_{W_0} : T \hookrightarrow T^2 \oplus R_0. \\
 \left. \begin{aligned} x &\in R_0, \quad F_{W_{-1}(\mu)}(x) = 0, \\ x &= F_{W_0}(y), \quad 0 = Ux = F_{W_0}(Uy) \end{aligned} \right\} &\implies Uy = 0 \implies y = x = 0.
 \end{aligned}$$

$F_{W_{-1}(\mu)}|_{R_0}$ injective, $\dim_{\mathbb{F}} R_0 = \dim_{\mathbb{F}} R_1 = 2 \implies F_{W_{-1}(\mu)}|_{R_0}$ is an isomorphism. Thus the two-handle map is an isomorphism on reduced Floer homology. $\square$

The one- and three-handle maps are

$$\begin{aligned} \mathrm{HF}^-(Y \# (S^2 \times S^1)) &\cong \mathrm{HF}^-(Y) \otimes_{\mathbb{F}} H_*(S^1), & F_1(x) &= x \otimes \theta^+, \\ F_3(x \otimes \theta^+) &= 0, & F_3(x \otimes \theta^-) &= x. \end{aligned}$$

Proposition 2.2.5 (Product formula, [21]). *Let $b_1(Y) \geq 2$ and $c_1(\mathfrak{s}) \notin \mathrm{Tor} H^2(Y)$. For $\pi: Y \times S^1 \rightarrow Y$,*

$$\begin{aligned} b^+(Y \times S^1) &= b^-(Y \times S^1) = b_1(Y), \\ \mathrm{SW}_{Y \times S^1, \pi^* \mathfrak{s}} &\equiv \dim_{\mathbb{F}} \mathrm{HM}_{\mathrm{red}}(Y, \mathfrak{s}) \equiv \dim_{\mathbb{F}} \mathrm{HF}_{\mathrm{red}}(Y, \mathfrak{s}) \pmod{2}, \\ \mathfrak{t}|_Y = \mathfrak{s}, \quad \mathfrak{t} \neq \pi^* \mathfrak{s} &\implies \mathrm{SW}_{Y \times S^1, \mathfrak{t}} \equiv 0 \pmod{2}. \end{aligned}$$

Proof. Choose dual bases $A_i \in H_2(Y; \mathbb{R})$, $a_i \in H_1(Y; \mathbb{R})$, and put $B_i = a_i \times S^1$.

$$A_i \cdot A_j = B_i \cdot B_j = 0, \quad A_i \cdot B_j = \delta_{ij} \implies \mathcal{Q}_{Y \times S^1} \cong b_1(Y)H \implies b^+ = b^- = b_1(Y).$$

$$Y \times S^1 = \mathrm{cl}(Y \times I), \quad F_{Y \times I} = \mathrm{id},$$

$$\sum_{\mathfrak{t}|_Y = \mathfrak{s}} \mathrm{SW}_{Y \times S^1}(\mathfrak{t}) \xrightarrow{2,2,2} \mathrm{Lef}(\mathrm{id}_{\mathrm{HM}_{\mathrm{red}}(Y, \mathfrak{s})}) \equiv \dim_{\mathbb{F}} \mathrm{HF}_{\mathrm{red}}(Y, \mathfrak{s}) \pmod{2}.$$

$$\eta \in H^1(S^1; \mathbb{Z}), \quad a \in H^1(Y; \mathbb{Z}), \quad c_1(\mathfrak{t}) = c_1(\pi^* \mathfrak{s}) + 2a \smile \eta,$$

$$\{[\gamma_i]\}_{i=1}^{b_1(Y)} \text{ basis of } H_1(Y; \mathbb{Z})/\mathrm{Tor}, \quad \mathrm{SW}_{Y \times S^1}(\mathfrak{t}) \neq 0,$$

$$\mathrm{SW}_{Y \times S^1}(\mathfrak{t}) \neq 0 \xrightarrow{\mathrm{SW-A}} 0 = |\langle c_1(\mathfrak{t}), [\gamma_i \times S^1] \rangle| = 2|\langle a, [\gamma_i] \rangle| \quad (1 \leq i \leq b_1(Y)) \implies a = 0.$$

Thus only $\pi^* \mathfrak{s}$ contributes, proving all three formulas. $\square$

2.2.1. The Eleven-Handle Cobordism.

Proposition 2.2.6 (Pretzel surgery identities, [20]). *If 0 denotes the unknot, then*

$$\begin{aligned} S_{p,-1}^3(K_n \cup \mu) &\cong S_{p+1}^3(K_n), & S_{p,-1}^3(K_n \cup \gamma) &\cong S_p^3(K_{n+2}), \\ S_{p,0}^3(K_n \cup \gamma) &\cong S_p^3(0) \# (S^2 \times S^1), & S_{-1}^3(K_n) &\cong S_1^3(K_{n+7}), \\ S_{-1}^3(K_{-8}) &\cong S_1^3(K_{-1}) = S_1^3(6_1). \end{aligned}$$

Proof. The four Kirby transformations are blow-down, splitting, and dualizable-pattern slides.

$$(K_n \cup \mu; p, -1) \xrightarrow{\mu\text{-blow-down}} (K_n; p+1)$$

$$(K_n \cup \gamma; p, -1) \xrightarrow{\gamma\text{-blow-down}} (K_{n+2}; p)$$

$$(K_n \cup \gamma; p, 0) \xrightarrow{\text{ribbon split}} (U; p) \sqcup (U; 0)$$

$$(K_n; -1) \xrightarrow{\text{dualizable-pattern slides}} (K_{n+7}; +1).$$

Since $S_0^3(U) \cong S^2 \times S^1$, the four rows give the asserted fillings. $\square$

Proposition 2.2.7 (The self-cobordism, [20]). *The eleven two-handles assemble as*

$$S_{-1}^3(K_n) \xrightarrow{2\mu} S_1^3(K_n) \cong S_{-1}^3(K_{n-7}) \xrightarrow{2\mu} S_1^3(K_{n-7}) \cong S_{-1}^3(K_{n-14}) \xrightarrow{7\gamma} S_{-1}^3(K_n).$$

$$n = -8 \xrightarrow{2,2,6} W : S_1^3(6_1) \longrightarrow S_1^3(6_1).$$

Let X be the identity closure and X^σ the closure by σ .

Proof. Each meridional block changes the framing twice, and each γ -handle raises the pretzel parameter by two.

$$S_{-1}^3(K_j) \xrightarrow{\mu} S_0^3(K_j) \xrightarrow{\mu} S_1^3(K_j) \cong S_{-1}^3(K_{j-7}),$$

$$K_{n-14} \xrightarrow{7\gamma} K_{n-14+14} = K_n, \quad n = -8 \xrightarrow{2,2,6} \partial_- W \cong \partial_+ W \cong S_1^3(6_1).$$

Thus the eleven handles form the stated self-cobordism and its two closures. $\square$

Theorem 2.2.8 (Levine–Lidman–Piccirillo, [20]). *Write*

$$\mathrm{HF}_{\mathrm{red}}(S_1^3(6_1)) = \mathbb{F}a \oplus \mathbb{F}b, \quad \sigma_*(a) = b, \quad \sigma_*(b) = a.$$

Exactly two Spin^c -structures $\mathfrak{t}, \mathfrak{t}'$ on W act nontrivially on $\mathrm{HF}_{\mathrm{red}}$. For good bases $(a, b), (a', b')$ at the ends,

$$F_{W,\mathfrak{t}}^{\mathrm{HF}}(a) = a', \quad F_{W,\mathfrak{t}}^{\mathrm{HF}}(b) = 0, \quad F_{W,\mathfrak{t}'}^{\mathrm{HF}}(a) = 0, \quad F_{W,\mathfrak{t}'}^{\mathrm{HF}}(b) = b',$$

$$\{a', b'\} = \{a, b\}, \quad \mathfrak{v} \notin \{\mathfrak{t}, \mathfrak{t}'\} \implies F_{W,\mathfrak{v}}^{\mathrm{HF}}|_{\mathrm{HF}_{\mathrm{red}}} = 0.$$

Proof. By Figure 6, σ_* exchanges the images of the two complementary projections computed by the eleven-handle reduced triangles.

$$\begin{array}{c} \mathbb{F}a \oplus \mathbb{F}b \xrightarrow{F_{W,\mathfrak{t}}} \mathbb{F}a' \oplus \mathbb{F}b' \xrightarrow{\sigma_*} \mathbb{F}b' \oplus \mathbb{F}a' \\ \xrightarrow{F_{W,\mathfrak{t}'}} \end{array}$$

$$[F_{W,\mathfrak{t}}]_{(a,b)}^{(a',b')} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad [F_{W,\mathfrak{t}'}]_{(a,b)}^{(a',b')} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad [\sigma_*] = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Thus precisely the two stated Spin^c -summands survive on $\mathrm{HF}_{\mathrm{red}}$. $\square$

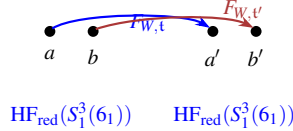


FIGURE 14. The two nonzero maps of the eleven-handle self-cobordism.

Theorem 2.2.9 (Topology and exoticity of the closures, [20]). *For $\Lambda = \mathbb{Z}[t, t^{-1}]$,*

$$\pi_1(X) \cong \pi_1(X^\sigma) \cong \mathbb{Z}, \quad \lambda_X \cong \lambda_{X^\sigma} \cong 2\langle 1 \rangle \oplus 9\langle -1 \rangle \quad \text{over } \Lambda,$$

$$X \cong_{\mathrm{homeo}} X^\sigma \cong_{\mathrm{homeo}} (S^1 \times S^3) \# 2\mathbb{CP}^2 \# 9\overline{\mathbb{CP}}^2, \quad X \not\cong_{\mathrm{sm}} X^\sigma.$$

Proof. Compute the closure groups, equivariant forms, and Floer traces from the eleven handles.

$$\begin{aligned}
\pi_1(W) &= \frac{\pi_1(S_{-1}^3(K_{-8}))}{\langle\langle \mu \rangle\rangle} = 1, & \pi_1(X) &= \pi_1(X^\sigma) = \langle t \mid \rangle \cong \mathbb{Z}. \\
Q_W &\cong 7\langle -1 \rangle \oplus 2 \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix} \cong 2\langle 1 \rangle \oplus 9\langle -1 \rangle, & b^+(W) &= b^+(X) = b^+(X^\sigma) = 2. \\
H_*(S_1^3(6_1)) &\cong H_*(S^3), & 0 &\longrightarrow \bigoplus_{k \in \mathbb{Z}} H_2(t^k W) \longrightarrow H_2(\tilde{X}) \longrightarrow 0, \\
H_2(\tilde{X}) &\cong H_2(W) \otimes_{\mathbb{Z}} \Lambda \cong H_2(\tilde{X}^\sigma), & t^i H_2(W) \cdot t^j H_2(W) &= 0 \quad (i \neq j), \\
\lambda_X &\cong \lambda_{X^\sigma} \cong Q_W \otimes_{\mathbb{Z}} \Lambda \cong 2\langle 1 \rangle \oplus 9\langle -1 \rangle. \\
\left. \begin{aligned} \pi_1(X) &= \pi_1(X^\sigma) = \mathbb{Z}, \\ \lambda_X &\cong \lambda_{X^\sigma}, \\ \lambda_X &\text{ is odd,} \\ \text{KS}(X) &= \text{KS}(X^\sigma) = 0 \end{aligned} \right\} & \xrightarrow{[11, 20]} X \cong_{\text{homeo}} X^\sigma \cong_{\text{homeo}} (S^1 \times S^3) \# 2\mathbb{CP}^2 \# 9\overline{\mathbb{CP}}^2. \\
Y := S_1^3(6_1), & H^1(Y) = H^2(Y) = 0 \implies \text{Spin}^c(X) \xrightarrow{\cong} \text{Spin}^c(W) \xleftarrow{\cong} \text{Spin}^c(X^\sigma). \\
\begin{array}{c|cc|cc} (a', b') & \text{Tr}(F_{W, \mathfrak{t}}^{\text{HF}}) & \text{Tr}(F_{W, \mathfrak{t}'}^{\text{HF}}) & \text{Tr}(\sigma_* F_{W, \mathfrak{t}}^{\text{HF}}) & \text{Tr}(\sigma_* F_{W, \mathfrak{t}'}^{\text{HF}}) \\ \hline (a, b) & 1 & 1 & 0 & 0 \\ (b, a) & 0 & 0 & 1 & 1 \end{array} & \text{by Theorem 2.2.8.} \\
\mathfrak{v} \notin \{\mathfrak{t}, \mathfrak{t}'\} & \implies \text{Tr}(F_{W, \mathfrak{v}}^{\text{HF}}) = 0, \\
\left. \begin{aligned} &\text{the displayed trace table,} \\ b^+(X) &= b^+(X^\sigma) = 2, \\ \text{Tr}(F_{W, \mathfrak{v}}^{\text{HF}}) &= 0 \quad (\mathfrak{v} \notin \{\mathfrak{t}, \mathfrak{t}'\}) \end{aligned} \right\} & \xrightarrow{2.2.2} \{\#\text{supp } \Phi_X, \#\text{supp } \Phi_{X^\sigma}\} = \{2, 0\} \implies X \not\cong_{\text{sm}} X^\sigma.
\end{aligned}$$

Thus X and X^σ form the stated exotic pair. □

Part 2. Exercises

3. EXERCISES

Exercises marked (*) are recommended; (!) marks a suggested starting point.

Exercise 3.0.1 (1.1.4). (*) Let X, X' be simply connected, closed, oriented, indefinite, and homeomorphic.

- (1) By Theorem 1.1.2, show that X, X' are smoothly h -cobordant.
- (2) Alternatively, assuming X, X' are smoothly h -cobordant, show that

$$X \#_k (S^2 \times S^2) \#_\ell (S^2 \widetilde{\times} S^2) \cong_{\text{sm}} X' \#_k (S^2 \times S^2) \#_\ell (S^2 \widetilde{\times} S^2)$$

for some k, ℓ . Hint: trade away the 1- and 4-handles.

Exercise 3.0.2 (1.1.5). (!) Show that surgery on $S^2 \times \text{pt} \subset X \# (S^2 \times S^2)$ returns X .

Exercise 3.0.3 (1.1.7). (*, !) Suppose

$$f: X \xrightarrow{\cong_{\text{homeo}}} X', \quad g: X \# \overline{\mathbb{CP}}^2 \xrightarrow{\cong_{\text{sm}}} X' \# \overline{\mathbb{CP}}^2, \quad g_* = f_* \oplus \text{id}.$$

Show that $X \# \overline{\mathbb{CP}}^2$ contains an exotic S^2 .

Exercise 3.0.4 (1.1.12). (*)

- (1) Show that $K3$ contains no square-2 torus.
- (2) Show that $K3 \not\cong X_1 \# X_2$ when neither X_i is homeomorphic to S^4 . Use

$$Q_{K3} \cong 2(-E_8) \oplus 3H, \quad H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

spin, simple connectivity, and Donaldson diagonalization.

Exercise 3.0.5 (1.1.12). Compute the Seiberg–Witten invariants of T^4 .

Exercise 3.0.6 (1.1.12). Let X be closed and symplectic, so $b^+(X) \geq 1$.

- (1) (!) Show that the connected sum of two closed symplectic four-manifolds is not symplectic.
- (2) Show that $X \# Y$ is not symplectic whenever

$$\exists f: \pi_1(Y) \longrightarrow G, \quad 1 < |G| < \infty, \quad f \neq 1.$$

Exercise 3.0.7 (1.1.14). (*) For $X_n(K) := B^4 \cup_K h_n^2$, let K be a negative torus knot. Show

$$\text{SW}_M \neq 0, \quad n \geq 0 \implies X_n(K) \not\rightarrow M.$$

Hint: use the blow-up formula.

Exercise 3.0.8 (1.1.15). (*) Put

$$d(\mathfrak{s}) := \frac{c_1(\mathfrak{s})^2 - 3\sigma(X) - 2\chi(X)}{4}.$$

Replace one natural $-E_8 \subset K3$, with boundary $S_{-1}^3(T_{-2,3})$, by $X_{-1}(T_{-2,3})$. Show that the resulting manifold has no basic classes. You may use

$$Q_{K3 \setminus ((-E_8) \sqcup (-E_8))} \cong 3 \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix},$$

with each summand represented by a square-(-2) sphere and a square-0 torus.

Exercise 3.0.9 (1.2.2). (*)

- (1) Show that $X_g(T)$ is determined by $g_*(1, 0, 0)$.
- (2) For $T = K \times S^1 \subset S^3 \times S^1$, show

$$(S^3 \times S^1)_{p,q,0}(T) \cong S_{p/q}^3(K) \times S^1.$$

Exercise 3.0.10 (1.2.4). For $X = (S^1 \times D^2) \times S^1$ and $T_n = J_n \times S^1$, show

$$X_{1,m,0}(T_n) = (S^1 \times D_{1/m}^2(J_n)) \times S^1 \cong (S^3 \setminus Tw_{n,-m}) \times S^1.$$

By Figures 2 and 2, verify the identification.

Exercise 3.0.11 (1.2.5). For an unknotted torus $T \subset X$, show

$$X_{1,q,r}(T) \cong_{\text{sm}} X \quad (q, r \in \mathbb{Z}).$$

Alternatively, find $X, T, (p, q, r) \neq \pm(1, 0, 0)$ such that X and $X_{p,q,r}(T)$ have isomorphic fundamental groups and intersection forms.

Exercise 3.0.12 (1.2.12). Show that knot surgery preserves homology and the intersection form.

Exercise 3.0.13 (1.2.12). Let $T \subset X$ be π_1 -null. Show

$$\pi_1(X) = 1, \quad \pi_1(X(T, K)) = 1$$

for every knot $K \subset S^3$.

Exercise 3.0.14 (1.2.14). (Hard without Lefschetz fibrations) Show that a torus fiber $T \subset E(2) = K3$ is π_1 -null.

Exercise 3.0.15 (1.2.15). Show that every knot surgery is a sequence of torus surgeries. Hint: generalize Exercise 3.0.10.

Exercise 3.0.16 (1.2.19). Let $S \subset K3$ be the twist torus from Exercise 3.0.10, and let S' be the unknotted torus.

(1) Show

$$\pi_1(K3 \setminus S') \cong \pi_1(K3 \setminus S) \cong \mathbb{Z}.$$

(2) Compute the equivariant intersection forms of $K3 \setminus S'$ and $K3 \setminus S$.

Exercise 3.0.17 (1.2.21). If $T = \partial(S^1 \times D^2) \subset X$, show

$$X(T, K) \cong_{\text{sm}} X$$

for every knot K .

Exercise 3.0.18 (1.2.23). Show that Σ_K is obtained by knot surgery along the rim torus

$$\mu \times S^1, \quad \mu = \text{a meridian of } \{0\} \times I.$$

Exercise 3.0.19 (1.2.24). Determine the genera, orientability, and self-intersections for which regluing $v\Sigma$ can change the smooth structure of X . Hint: use the mapping class groups of S^1 -bundles over surfaces.

Exercise 3.0.20 (2.1.4). (*) For a slice disk $D_K \subset B^4$, let

$$C_{D_K} := (B^4 \setminus \text{int}(vD_K)) \cup_{\mu_K} h_{-1}^2.$$

(1) Show $\partial C_{D_K} = S_{+1}^3(K)$.

(2) Show $C_{D_K} \simeq \text{pt}$. Hint: Hurewicz and Whitehead.

(3) Show that C_{D_K} is also obtained by surgery on D_K .

Exercise 3.0.21 (2.1.5). Let ι be a strong inversion of K , and assume

$$\iota(z, e^{i\theta}) = (\bar{z}, e^{-i\theta})$$

on an invariant neighborhood $D^2 \times S^1$ of K . Show that $\iota|_{S^3 \setminus K}$ extends to an involution of ∂C_{D_K} .

Exercise 3.0.22 (2.1.8). (!) If $\phi: Y \rightarrow Y$ extends to $\Phi: W \rightarrow W$, show

$$\phi_* r(W, \mathfrak{t}) = r(W, \Phi_* \mathfrak{t}).$$

Exercise 3.0.23 (2.1.9). (!) Prove Theorem 2.1.9 when C has one 1-handle and one 2-handle. Hint: double C° along ∂C and use Theorem 2.1.6.

Exercise 3.0.24 (2.1.11). (*) Prove Theorem 2.1.11 for τ .

Exercise 3.0.25 (2.1.26). (*,!) Let $\mathcal{H}$ be a handle diagram of B^4 , let $K \subset \partial \mathcal{H}$ be unknotted and split from its 1-handles, and let ι restrict to a strong inversion of K .

- (1) Show that K bounds the evident disk $D_K \subset B^4$.
- (2) Show that ι induces

$$\tau': \partial E(D_K) \longrightarrow \partial E(D_K).$$

- (3) Show

$$\tau' \text{ extends diffeomorphically over } E(D_K) \implies \tau'' \text{ extends diffeomorphically over } C_{D_K}.$$

- (4) Deduce

$$(C_{D_K}, \tau'') \text{ is a cork} \implies (D_K, \iota) \text{ is a surface cork.}$$

Exercise 3.0.26 (2.1.27). By Figures 11 and 8, show

$$(C_{D_K}, \tau'') \cong (P, \tau) \implies (D_K, \iota) \text{ is a surface cork.}$$

Exercise 3.0.27 (2.1.23). By Figure 10, verify that the two marked slides give Figure 8.

Exercise 3.0.28 (2.1.24). For $\pi: \Sigma_2(B^4, D_P) \rightarrow B^4$, verify

- (1) $\Sigma_2(B^4, D_P) \cong P$.
- (2) The boundary involution is τ .
- (3) The covering involution is ρ .
- (4) $\iota \circ \pi = \pi \circ \tau$ on ∂P .

Exercise 3.0.29 (2.2.3). (*,!) Show

$$S^3 \subset X \text{ nonseparating} \implies \text{SW}_X \equiv 0.$$

Exercise 3.0.30 (2.2.4). (*) Using

$$\text{HF}^-(S_1^3(K_n)) \cong \mathbb{F}[U] \oplus (\mathbb{F}[U]/U)^2, \quad \text{HF}^-(S_0^3(K_n)) \cong \mathbb{F}[U]^2 \oplus (\mathbb{F}[U]/U)^2,$$

and the completed surgery triangle, show

$$F_{W^{-1}(\mu)}: \text{HF}_{\text{red}}(S_0^3(K_n)) \xrightarrow{\cong} \text{HF}_{\text{red}}(S_1^3(K_n)).$$

Exercise 3.0.31 (2.2.5). Let $b_1(Y) \geq 2$.

- (1) (*) Show

$$b^+(Y \times S^1) = b_1(Y).$$

- (2) (!) For non-torsion $\mathfrak{s} \in \text{Spin}^c(Y)$, show

$$\dim_{\mathbb{F}} \text{HF}_{\text{red}}(Y, \mathfrak{s}) \equiv \sum_{\mathfrak{t}|_{Y \times \{0\}} = \mathfrak{s}} \text{SW}_{Y \times S^1}(\mathfrak{t}) \pmod{2}.$$

- (3) Show that only $\pi^* \mathfrak{s} \in \text{Spin}^c(Y \times S^1)$ can contribute, and

$$\text{SW}_{Y \times S^1}(\pi^* \mathfrak{s}) \equiv \dim_{\mathbb{F}} \text{HF}_{\text{red}}(Y, \mathfrak{s}) \pmod{2}.$$

Exercise 3.0.32 (2.2.6). Verify

- (1) (!)

$$S_{p,-1}^3(K_n \cup \mu) \cong S_{p+1}^3(K_n).$$

- (2)

$$S_{p,-1}^3(K_n \cup \gamma) \cong S_p^3(K_{n+2}).$$

(3) (*)

$$S_{p,0}^3(K_n \cup \gamma) \cong S_p^3(0) \# (S^2 \times S^1).$$

(4)

$$S_{-1}^3(K_n) \cong S_1^3(K_{n+7}), \quad S_{-1}^3(K_{-8}) \cong S_1^3(6_1).$$

Exercise 3.0.33 (2.2.9). (*) Show

$$\pi_1(X) \cong \pi_1(X^\sigma) \cong \mathbb{Z}.$$

Exercise 3.0.34 (2.2.9). Compute the equivariant intersection forms of X, X^σ , and show that they agree with that of

$$(S^1 \times S^3) \# 2\mathbb{CP}^2 \# 9\overline{\mathbb{CP}}^2.$$

Exercise 3.0.35 (2.2.9). Use Freedman–Quinn classification to show

$$X \cong_{\text{homeo}} X^\sigma \cong_{\text{homeo}} (S^1 \times S^3) \# 2\mathbb{CP}^2 \# 9\overline{\mathbb{CP}}^2, \quad X \not\cong_{\text{sm}} X^\sigma.$$

REFERENCES

- [1] R. İ. Baykur and N. Sunukjian, *Round handles, logarithmic transforms and smooth 4-manifolds*, J. Topol. **6** (2013), no. 1, 49–63.
- [2] S. Boyer, *Simply-connected 4-manifolds with a given boundary*, Trans. Amer. Math. Soc. **298** (1986), no. 1, 331–357.
- [3] A. Conway and M. Powell, *Embedded surfaces with infinite cyclic knot group*, Geom. Topol. **27** (2023), no. 2, 739–821.
- [4] I. Dai, M. Hedden, and A. Mallick, *Corks, involutions, and Heegaard Floer homology*, J. Eur. Math. Soc. (JEMS) **25** (2023), no. 6, 2319–2389.
- [5] S. K. Donaldson, *An application of gauge theory to four-dimensional topology*, J. Differential Geom. **18** (1983), no. 2, 279–315.
- [6] S. K. Donaldson, *Irrationality and the h-cobordism conjecture*, J. Differential Geom. **26** (1987), no. 1, 141–168.
- [7] R. Fintushel and R. J. Stern, *Surfaces in 4-manifolds*, Math. Res. Lett. **4** (1997), no. 6, 907–914.
- [8] R. Fintushel and R. J. Stern, *Surfaces in 4-manifolds: addendum*, preprint, 2005, arXiv:math/0511707.
- [9] R. Fintushel and R. J. Stern, *Knots, links, and 4-manifolds*, Invent. Math. **134** (1998), no. 2, 363–400.
- [10] M. H. Freedman, *The topology of four-dimensional manifolds*, J. Differential Geom. **17** (1982), no. 3, 357–453.
- [11] M. H. Freedman and F. Quinn, *Topology of 4-Manifolds*, Princeton Mathematical Series **39**, Princeton University Press, Princeton, NJ, 1990.
- [12] M. Furuta, *Monopole equation and the 11/8-conjecture*, Math. Res. Lett. **8** (2001), no. 3, 279–291.
- [13] C. McA. Gordon and J. Luecke, *Knots are determined by their complements*, J. Amer. Math. Soc. **2** (1989), no. 2, 371–415.
- [14] N. R. Hoffman and N. S. Sunukjian, *Null-homologous exotic surfaces in 4-manifolds*, Algebr. Geom. Topol. **20** (2020), no. 5, 2677–2685.
- [15] T. Kato, N. Nakamura, and K. Yasui, *The simple type conjecture for mod 2 Seiberg–Witten invariants*, J. Eur. Math. Soc. (JEMS) **25** (2023), no. 12, 4869–4877.
- [16] H. J. Kim, *Modifying surfaces in 4-manifolds by twist spinning*, Geom. Topol. **10** (2006), 27–56.
- [17] P. B. Kronheimer, T. S. Mrowka, P. S. Ozsváth, and Z. Szabó, *Monopoles and lens space surgeries*, Ann. of Math. (2) **165** (2007), no. 2, 457–546.
- [18] M. Kreck, *h-cobordisms between 1-connected 4-manifolds*, Geom. Topol. **5** (2001), 1–6.
- [19] P. B. Kronheimer and T. S. Mrowka, *The genus of embedded surfaces in the projective plane*, Math. Res. Lett. **1** (1994), no. 6, 797–808.
- [20] A. S. Levine, T. Lidman, and L. Piccirillo, *New constructions and invariants of closed exotic 4-manifolds*, preprint, 2023, arXiv:2307.08130.
- [21] Tye Lidman and Lisa Piccirillo, *Techniques in exotica*.
- [22] P. Lisca and G. Matić, *Stein 4-manifolds with boundary and contact structures*, Topology Appl. **88** (1998), no. 1–2, 55–66.
- [23] C. Manolescu, *Pin(2)-equivariant Seiberg–Witten Floer homology and the Triangulation Conjecture*, J. Amer. Math. Soc. **29** (2016), no. 1, 147–176.
- [24] T. E. Mark, *Knotted surfaces in 4-manifolds*, Forum Math. **25** (2013), no. 3, 597–637.
- [25] G. Meng and C. H. Taubes, *SW = Milnor torsion*, Math. Res. Lett. **3** (1996), no. 5, 661–674.
- [26] J. Milnor, *On manifolds homeomorphic to the 7-sphere*, Ann. of Math. (2) **64** (1956), no. 2, 399–405.
- [27] J. W. Morgan, T. S. Mrowka, and Z. Szabó, *Product formulas along T^3 for Seiberg–Witten invariants*, Math. Res. Lett. **4** (1997), no. 6, 915–929.
- [28] J. W. Morgan, *The Seiberg–Witten Equations and Applications to the Topology of Smooth Four-Manifolds*, Mathematical Notes **44**, Princeton University Press, Princeton, NJ, 1996.

- [29] A. Némethi and L. I. Nicolaescu, *Seiberg–Witten invariants and surface singularities*, Geom. Topol. **6** (2002), 269–328.
- [30] L. I. Nicolaescu, *Notes on Seiberg–Witten Theory*, Graduate Studies in Mathematics **28**, American Mathematical Society, Providence, RI, 2000.
- [31] P. Ozsváth and Z. Szabó, *The symplectic Thom conjecture*, Ann. of Math. (2) **151** (2000), no. 1, 93–124.
- [32] S. Smale, *On the structure of manifolds*, Amer. J. Math. **84** (1962), 387–399.
- [33] N. S. Sunukjian, *A note on knot surgery*, J. Knot Theory Ramifications **24** (2015), no. 9, 1520003, 5 pp.
- [34] N. S. Sunukjian, *Surfaces in 4-manifolds: concordance, isotopy, and surgery*, Int. Math. Res. Not. IMRN (2015), no. 17, 7950–7978.
- [35] C. H. Taubes, *The Seiberg–Witten invariants and symplectic forms*, Math. Res. Lett. **1** (1994), no. 6, 809–822.
- [36] C. T. C. Wall, *On simply-connected 4-manifolds*, J. London Math. Soc. **39** (1964), 141–149.
- [37] E. Witten, *Monopoles and four-manifolds*, Math. Res. Lett. **1** (1994), no. 6, 769–796.